

Specification

Specification Book

Scope

Applicable Document of Part 1
Common Electrical Specification for Supplier

Basic DID Definition Specification

Range of Validity

VinFast Vehicles

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Specification

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General Information

1.1 Language

This specification is written in English.

1.2 Change History

Version	Chapter	Brief Description	Date	Edited By
1.0.0	All	# Initial Version	2019.10.14	Nguyen Hoang Long
1.0.0	All	# Pdf -> Word Conversion and further update w.r.to C-SUV	2020.01.02	Rasul Nadaf
1.0.1	All	# Change DID 0xF18A from 5 byte ASCII to 4 byte BCD # Changed usage of DID 0xF190 as Mandatory for EMS, BCM & MHU # Changed usage of DID 0xF102 (was previously mandatory) # Changed parameter example of DID 0xF103, - 7th Byte low nibble is needed - 8th byte marked as ASCII Character "V" to match with Team centre # Changed usage of DID 0xF111 marked for BCM/XGW and MHU # Changed usage of DID 0xF112 marked for BCM/XGW and MHU # Changed usage of DID 0xF113 marked for BCM/XGW and MHU # Changed DID 0xF11A name from PT to Powertrain # Changed DID 0xF11B name from TC to Chassis # Deleted DID 0xF11F as its not applicable # Changed usage of DID 0xFD01 (was previously mandatory) # Changed usage of DID 0xF194(was previously mandatory) # Added below New DID as per VF35 architecture - DID 0xF125 ECU ADAS CAN dbc file version - DID 0xF126 ECU Diagnostic CAN dbc file version - DID 0xF127 ECU FA CAN dbc file version - DID 0xF128 ECU FLEXRAY FIBEX file version - DID 0xF129 ECU LIN3 LDF file version	2021.04.25	Rasul Nadaf
1.1	3	1) Specification document name changed to VFDSVAEEP0002. 3) Added below New DID as per architecture: - DID 0xF120 Time Stamp - Global snapshot - DID 0xF121 Odometer - Global snapshot - DID 0xF122 ECU Supply Voltage - Global snapshot	2021.10.22	Daniel Canelopol

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		- DID 0xF123 Ignition status - Global snapshot - DID 0xF18C ECUSerialNumberDataIdentifier - DID DD01 Cabin Temperature - DID DD02 Outside Temperature - DID DD03 ECU Temperature - DID DD04 ECU Voltage - DID D00D Vehicle Communication Bus State - DID DD06 Engine status - Global snapshot - DID DD07 GPS Time - Global snapshot - DID DD08 Engine RPM - Global snapshot - DID DD09 Vehicle speed - Global snapshot - DID F18F RegulationXSoftware Identification Number (RxSWIN) - DID D020 Internal Fail Code - DID 4884 Internal Fault Data - DID F800 VIN2 - DID F801 VIN3 - DID F802 VIN4 - DID F141 Hardware Number Revision Level - DID F142 Calibration Software Number Revision Level - DID F143 Purchase Part Number Revision Level - DID F144 Application Software Part Number Revision Level - DID F145 Basic Software Part Number Revision Level - DID F146 HMI Software Part Number Revision Level - DID F148 Software Part Number Revision Level 4) Removed DID F18A.		Daniel Canelopol
1.2	3	- Format of the following DID's has been changed to match Team Centre Part Number – F188, F191, F102, F103, F104, F105, F106. - Flash Boot Loader defined for all DID's. - CAN signals referenced for Snapshot DID's	2021.11.29	Daniel Canelopol
1.3	3	- F18C DID – Write once in Bootloader added - D005 DID changed to DD05 - F120 DID name change to Global Time - F18F DID – parameter details added - D020 DID – parameter details updated 016C DID – new DID added - D05B DID – new DID added - D06B DID – new DID added - F190 DID – Read in Bootloader changed to N/A	2022.01.28	Daniel Canelopol
1.4	3.2 3.14 3.15	- DID F188 correctly defined - DID F111 no value changed from 00 to FF CRC check updated - DID F112 no value changed from 00 to FF CRC check updated	20/4/2022	Daniel Canelopol & Ashad Qureshi

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3.44	4)	DID DD05 Motor RPM size increase to 4bytes and note added resolution corrected and right justified		
3.31	5)	DID FD01 Note added		
3.64	6)	DID D06B Usage clarified		
3.46	7)	DID DD07 corrected size to 6 bytes to match F120		
3.48	8)	DD09 typo corrected from 1bit=255 to 1 byte =255km/h		
3.52	9)	Change the requirement of additional VIN 0xF800 storage from M to M/C and a note added		
3.53	10)	Change the requirement of additional VIN 0xF801 storage from M to M/C and a note added		
3.54	11)	Change the requirement of additional VIN 0xF802 storage from M to M/C and a note added		
3.55	12)	Added a 5 th VIN 0x0803 for ECU which ties the calibration to a vehicle and note added.		
3.65	13)	DID D05B changed from mandatory to conditional		
3.18	14)	DID 0xF113 DID length corrected		
2.2	15)	Requirement clarified ref addressing of DID to say all DID support physical and functional addressing		
3.34	16)	DID 0xF110 Removed the ref ECU name added help text for it		
3.35	17)	Correction to the DID length F120		
3.36	18)	DID F121 Note added		
3.42	19)	DD03 ECU temperature clarification note added		
3.5	20)	F190 DID size correctly specified to 24 bytes		
3.33	21)	F194 notes added		
3.40	22)	DD01 Use case clarified		
3.41	23)	DD02 Use case clarified		
3.49	24)	D00D DID clarified and notes added.		
3.51	25)	DID4884 clarified and type changed from hex to packeted		
3.63	26)	DID 0x016C Changed the DID from unsigned to state encoded and updated table		
3.37	27)	F122 note added for capturing voltage from CAN		
3.66	28)	New DID F16A Current VOD DID for Category 1 ECU		
3.67	29)	New DID F16B Category 1 current and last stored VOD checksum		
3.68	30)	New DID F16C Category 1 Current VOD checksum		
3.69	31)	New DID F16D Category 1 and 2 VOD validation flag		
3.70	32)	0xEF00 VOD Validation State DID Category 2 ECU		
3.71	33)	0xEF01VOD Category 2 ECU compared checksum		
3.72	34)	0xEF02 Last stored VOD for Category 2 ECU		
3.73	35)	0xEF03 Last stored VOD for Category 1 ECU		
3.74	36)	0xEF04 VOD validation States for Category 1 ECU		
3.75	37)	0xEF05 Category 2 compare checksum		
4.2	38)	Added reference Document in table 5.		
3.39	39)	3.39 change the security for writing from the plant to supplier.		
3.16	40)	CDS added to the title		
3.18-				
3.25				
3.76				
3.77				
3.78				

VFDSVAEEP0002

	3.79 3.43 3.3 3.66	41) CAN DB file examples clarified 42) New DID F117 added 43) New DID F147 added 44) New DID 0804 added 45) New DID 0805 added 46) DID D004 Note added 47) F18F clarified as to what it will report 48) Added new DID 0x0222 Valid Application		
1.4 1.4 1.4 1.4 1.4 1.5	3 3.82 3.30 3.83 All 3.40,3.41 3.48 3.37 3.38 3.5	0x016C DID – Updated to align with ODX 0x0179 DID introduced to check ECU control of the contactor 0xF10F DID updated scope Autosar Version number DID added Implementation updated on DIDS DD01 and DD02 requirement clarified DD09 updated F122 vehicle voltage resolution change to match CAN DB F123 Ignition Status DID size changed to match CAN DB F190 Note added regarding writing and the title corrected	24/06/22 12/7/2022 14/07/2022 15/08/2022	Steve Clarke/Ashad Qureshi
1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5	3.80 3.83 3.84 3.85 3.86 3.77 3.78 3.87 3.88 3.0 All 4.2 4.1 4.9, 4.10 4 4.49	Updated text On 0x0805 “Flash” 0x0806 will now be used for “Autosar Version Number” 0x0807 ODX template v DID added 0x0808 EOL template V DID added 0x0809 AS template V DID added DID name changed to F109 from F117 DID name changed to F147 from F149 LIN Node 4 added LIN Node 5 added Info added Requirement to support Functional addressing for all DIDs which are greater than 8 bytes removed Updated DID 0xF18F RXSWIN to comply with ISO14229 2020 where DID is encoded all ASCII F188 usage redefined F104, F105 Usage redefined Added section 4 DID Standard for service ID 0x22 plus a short working description DID 0xD00D is now strikethrough – not to be used	23/09/2022 26/09/2022 29/09/2022 3/10/2022 3/10/2022 3/10/2022 11/10/2022 04/11/2022 13/12/2022 13/12/2022 03/01/23	Steve Clarke Ashad Qureshi Ashad Qureshi Ashad Qureshi Ashad Qureshi Steve Clarke A Qureshi A Qureshi Steve Clarke
1.6	4.2 4.3 4.6 4.7 4.9 4.10 4.11 4.12 4.65 4.14 4.15 4.16 4.17	DID F188 Part number length increased to 12 DID F18F Part number length increased to 13 DID F191 Part number length increased to 12 DID F102 Part number length increased to 12 DID F103 Part number length increased to 12 DID F104 Part number length increased to 12 DID F105 Part number length increased to 12 DID F106 Part number length increased to 12 DID D05B Part number length increased to 13 DID F111 Session clarified DID F112 Session clarified DID F108 session clarified DID F10A session Clarified	9/01/2023 07/02/2023	A Qurershish A Qureshi

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1.6	4.18	DID F113 Session Clarified	07/02/2023	
	All	Word mandatory removed.	08/03/2023	A Qureshi
	4.3	F18F Purchase part number added in byte 3-13	"	
1.6	4.3	Updated / Edited CANDB Messages and IDs	30/03/2023	S.Clarke
1.6	4.48	D00D removed not implemented and paragraph renumbered.	14/04/2023	A Qureshi
	4.82-	DIDs 0x0806, 0x0807, 0x0808 and 0x0809 DID made optional.		
	4.85			
	4.49	DID D00D removed and all other DIDs section renumbered.	25/04/2023	A Qureshi
	4.63	D05B requirement changed to make this readable in bootloader byte numbers corrected.	25/04/2023	A Qureshi

Table 1: Change History

Specification

2 Introduction

2.1 Coherence to other documents

The "Generic Diagnostic Specification-VFDSVAEEP0001-V1.5" refers to this document.

2.2 Terms and definitions

The following list of terms and definitions are used throughout this document with the given meaning.

All DIDs shall support physical and Functional addressing.

2.2.1 Abbreviations

Abbreviation	meaning
ECU	Electronic Control Unit
DID	Data Identifier
PID	Parameter Identifier
BOOT	Bootloader
Cvt	Convention
C	Conditional
M	Mandatory
O	Optional
U	User Conditional

Table 2 : List of abbreviations Definitions

2.2.2 Obligation of Requirement Wording

The following wording rules are used throughout this document.

Shall

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

Should

The word "should" is used for preferences or desires in implementation that are not strictly binding.

May

The word "may" is used to denote options that are left to the implementer. Security Level

The number and usage of Security Levels is described in the Generic_Diagnostic_Specification. Basically, the following Security Level exists:

Level	Request Seed	Send Key	Use case
1	0x01	0x02	Service
2	0x03	0x04	Manufacturing
3	0x05	0x06	Variant Coding
4	0x07	0x08	Flashing (ECU reprogramming)

Table 3 : Security Level

Level 0 means no security access is required.

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3 System Description

The DIDs detailed in this document are system DIDs to be implemented by ECU suppliers based on use case and to provide useful information such as Part numbers, vehicle status, vehicle environmental condition and snapshot data etc. The suppliers are therefore encouraged to use these DIDs to provide as much detail about their ECUs.

4 DID Standard for service ID 0x22

Data Byte No.	Parameter Name	Cvt	Value
0	Service positive response 0x62	M	
1	DID number upper Byte F1	M	
2	DID number lower Byte 88	M	
3	Data associated with DID	M	1bit to 1 Byte in length (0x01 to 0xFF)
4		C	
5		C	
6		C	
N+1		C	

The structure of all DIDs in this document will be as above where the first byte is byte 0 and will be the ISO14229 Service ID followed by the 2 byte DID value, the 3rd byte will be the first byte of any returned values when using service 0x22 i.e. 0x62 F1 88 [Data byte1].....[9]

Note attention must be paid to type of DID and Byte to be used.

ASCII is an ascii value that represents a physical key stroke Item e.g., ascii 0x30 = keystroke "0"

BCD is binary coded decimal which has following format 1 byte [upper nibble – lower nibble]

So can range from [0 to 9 – 0 to 9] so byte 3 may display 0x99 which = 99 decimal

SED = state encoded DID and has range from 0x00 to 0xFF where each state means something different e.g. 0x00 = OK, 0x01 = Fault, 0xFF means out of range.

A DID may be one type of value e.g. integer, BCD etc. or a mixture for example Part Numbers "ASCII,ASCII,ASCII,BCD,BCD,BCD,BCD"

Please see details for each DID listed, in the document and cross reference with the latest iso14229 standard.

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4.1 DID 0xF186 Active Diagnostic Session

Service	0x22			
DID	0xF186			
Short Description	Read Active Diagnostic Session			
Implementation	Application		Flash Boot Loader	
	Read only		Read	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Mandatory for all CAN ECUs and diagnostic Class III LIN node.			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
	3	sub-function = [DiagnosticSessionType		00 – FF
		defaultSession	M	01
		programmingSession	M	02
		extendedDiagnosticSession	M	03
		coding session	M/C	41
		Specific Session	O	40
		Specific Session	O	50
		]		
	Parameter:			
	- Active Diagnostic Session: 01 = Default Session 02 = Programming Session 03 = Extended Diagnostic Session 41 = Coding Session 40 = Specific session Bench Session 50 = Specific session Roller session			
	Note:			
	- These are the sub function parameters from Diagnostic Session Control service. - See also Generic_Diagnostic_Specification chapter “Diagnostic Services”. - All ECUs need to support sessions 10 01/81, 10 02/82 and 10 03/83 all other sessions are supported as case by case requirements.			

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4.2 DID 0xF188 Vehicle Manufacturer ECU Software Part Number

Service	0x22								
DID	0xF188								
Short Description	Software Part Number of the ECU defined by the OEM.								
Implementation	Application	Flash Boot Loader							
	Read(physical)	Updated during software download Not readable in Boot Loader							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECUs which are type 2 and type 3								
Data	Data Byte No.	Parameter Name	Cvt	Value					
		ECU Software Part Number [] = []							
	3	ECU Software Number Prefix Byte#0	M	ASCII					
	4	ECU Software Number Prefix Byte#1	M	ASCII					
	5	ECU Software Number Prefix Byte#2	M	ASCII					
	6	ECU Software Number Byte#0	M	BCD					
	7	ECU Software Number Byte#1	M	BCD					
	8	ECU Software Number Byte#2	M	BCD					
	9	ECU Software Number Byte#3	M	BCD					
	10	ECU SW Index Number	M	ASCII					
	11	ECU SW Index Number	M	ASCII					
	Parameter:								
	ECU Software Part Number from BOM/Teamcenter: e.g.: SOW12345678AA								
	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11
	53	4F	57	12	34	56	78	41	41
Note:									
- The OEM defines Vehicle Manufacture ECU Software Part Number, and it shall be embedded in the ECU Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Vehicle Manufacture ECU Software Part Number is stored. - Also see DID 0xF148 Vehicle Manufacturer ECU Software Part Number Revision Level.									
The ECUs will use this part number to report application and base software where both of these software components are combined. Where they are separate F104 and F105 shall be used.									

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4.3 DID 0xF18F Regulation-X Software Identification Number (RxSWIN)

Service	0x22																																														
DID	0xF18F																																														
Short Description	RegulationXSoftware Identification Number (RxSWIN)																																														
Implementation	Application										Flash Boot Loader																																				
	Read(physical only)										Updated during software download Not readable in PBL																																				
Security Level	Read										Write																																				
	Level 0										N/A																																				
Usage	Mandatory for ECUs that must adhere to RxSWIN Requirements																																														
Data	Data Byte No.	Parameter Name												Cvt	Value																																
	3 - 13	Purchase part Number []												M	ASCII																																
	14-15	Index												M	ASCII																																
	16	Space												M	ASCII																																
	17 -18	Revision Number												M	ASCII																																
Parameter: - The values should be aligned left. - The Content will be “ECU Purchase Part Number” then a index and “Space” then ECU Purchase Part Revision Number Example <i>ECU Purchase Part Number = EEP12345678AA plus index“Space” then “ECU Purchase Part Revision Number” = 01</i> <table><tr><td>Byte 3</td><td>Byte 4</td><td>Byte 5</td><td>Byte 6</td><td>Byte 7</td><td>Byte 8</td><td>Byte 9</td><td>Byte 10</td><td>Byte 11</td><td>Byte 12</td><td>Byte 13</td><td>Byte 14</td><td>Byte 15</td><td>Byte 16</td><td>Byte 17</td><td>Byte 18</td></tr><tr><td>45</td><td>45</td><td>50</td><td>31</td><td>32</td><td>33</td><td>34</td><td>35</td><td>36</td><td>37</td><td>38</td><td>41</td><td>41</td><td>20</td><td>30</td><td>31</td></tr></table> - Unused Bytes should be filled with space (0x20) - RxSWIN Identification number is a dedicated identifier, representing information about the type approval relevant software of the Electronic Control System contributing to the Regulation Number X type approval relevant characteristics of the vehicle. - Please refer to ECE-R156 & ISO14229-1 Annex C.																Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17	Byte 18	45	45	50	31	32	33	34	35	36	37	38	41	41	20	30	31
Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17	Byte 18																																
45	45	50	31	32	33	34	35	36	37	38	41	41	20	30	31																																
Note 1 RxSWIN DID will return ECU’s OEM Purchase part numbers related to software. The part numbers returned by an ECU will be relevant only to the ECU. E.g. VCU returns part numbers of VCU only																																															
Note 2 - This DID 0xF18F is all “ASCII” encoded the values are taken from 0xF103 and 0xF143 but must be converted into ASCII. - As Detailed in ISO14229 2020																																															

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4.4 DID 0xF18B ECU Manufacturing Date

Service	0x22																										
DID	0xF18B																										
Short Description	Read ECU Manufacturing Date																										
Implementation	Application	Flash Boot Loader																									
	Read (physical and functional addressing)	N/A																									
Security Level	Read	Write																									
	Level 0	N/A																									
Usage	Mandatory for all ECUs																										
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Manufacturing Date[] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Manufacturing Date - Year</td><td>M</td><td>(00 - 99) BCD</td></tr><tr><td>4</td><td>ECU Manufacturing Date - Month</td><td>M</td><td>(01 - 12) BCD</td></tr><tr><td>5</td><td>ECU Manufacturing Date - Day</td><td>M</td><td>(01 - 31) BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>			Data Byte No.	Parameter Name	Cvt	Value		ECU Manufacturing Date[] = [			3	ECU Manufacturing Date - Year	M	(00 - 99) BCD	4	ECU Manufacturing Date - Month	M	(01 - 12) BCD	5	ECU Manufacturing Date - Day	M	(01 - 31) BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																							
		ECU Manufacturing Date[] = [																									
	3	ECU Manufacturing Date - Year	M	(00 - 99) BCD																							
	4	ECU Manufacturing Date - Month	M	(01 - 12) BCD																							
	5	ECU Manufacturing Date - Day	M	(01 - 31) BCD																							
		]																									
	Parameter:																										
	Example ECU Manufacturing Date format: e.g: 2013.01.21																										
	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td></tr><tr><td>13</td><td>01</td><td>21</td></tr></table>			Byte3	Byte4	Byte5	13	01	21																		
Byte3	Byte4	Byte5																									
13	01	21																									
Note:																											
- Writing the date shall be done at the supplier site. - The ECU Manufacturing Date shows the production date of the ECU. - All values 00 means no ECU Manufacturing Date is stored.																											

Specification

4.5 DID 0xF190 Vehicle Identification Number (VIN)

Service	0x22 /0x2E																											
DID	0xF190																											
Short Description	Read / Write complete Vehicle Identification Number is (17 Characters) are shown																											
Implementation	Application		Flash Boot Loader																									
	Read/Write (physical addressing)		N/A																									
Security Level	Read		Write																									
	Level 0		Level1&2																									
Usage	Mandatory for all ECU's that supports diagnostics																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			3	Vehicle Identification Number High Byte#0	M	ASCII	...				19	Vehicle Identification Number Low Byte#16	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	3	Vehicle Identification Number High Byte#0	M	ASCII																								
	...																											
	19	Vehicle Identification Number Low Byte#16	M	ASCII																								
		]																										
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note: - The VIN write is a onetime write, once written it shall not be possible to overwrite it unless the application software is updated, in which case after the update of the software the VIN will need to be rewritten. - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER This signal will only provide the last 7 bytes of the VIN.																											

Specification

4.6 DID 0xF191 Vehicle Manufacturer ECU Hardware Part Number

Service	0x22																				
DID	0xF191																				
Short Description	Hardware Part Number of the ECU defined by the OEM.																				
Implementation	Application		Flash Boot Loader																		
	Read (physical addressing)		Read																		
Security Level	Read		Write																		
	Level 0		N/A																		
Usage	Mandatory																				
Data	Data Byte No.	Parameter Name	Cvt	Value																	
		ECU Software Part Number [] = []																			
	3	ECU Software Number Prefix Byte#0	M	ASCII																	
	4	ECU Software Number Prefix Byte#1	M	ASCII																	
	5	ECU Software Number Prefix Byte#2	M	ASCII																	
	6	ECU Software Number Byte#0	M	BCD																	
	7	ECU Software Number Byte#1	M	BCD																	
	8	ECU Software Number Byte#2	M	BCD																	
	9	ECU Software Number Byte#3	M	BCD																	
	10	ECU SW Index Number	M	ASCII																	
	11	ECU SW Index Number	M	ASCII																	
	Parameter:																				
	ECU Hardware Part Number from BOM/Teamcenter:																				
	e.g.: EEP12345678																				
	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
53	4F	57	12	34	56	78	41	41													
Note:																					
- The OEM defines Vehicle Manufacture ECU Hardware Part Number. - The Hardware Part Number must be written by the Supplier. - This DID is used to identify a unique combination of ECU Hardware and non-reprogrammable software (e.g. Boot Software). - If ECU hardware or the Boot Software Version is changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Hardware Part Number or Revision Level <u>must be changed</u>. - All Bytes 0x00 means no Vehicle Manufacture ECU Hardware Part Number is stored. - Also see DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level.																					

Specification

4.7 DID 0XF102 Vehicle Manufacturer ECU Calibration Part Number

Service	0x22																				
DID	0xF102																				
Short Description	Calibration Software Number of the ECU defined by the OEM																				
Implementation	Application	Flash Boot Loader																			
	Read(physical)	Updated during software download Not readable in Boot Loader																			
Security Level	Read	Write																			
	Level 0	N/A																			
Usage	Mandatory for ECUs where Calibration Part Number is to be reported separately																				
	Data Byte No.	Parameter Name	Cvt	Value																	
		ECU Software Part Number [] = []																			
	3	ECU Software Number Prefix Byte#0	M	ASCII																	
	4	ECU Software Number Prefix Byte#1	M	ASCII																	
	5	ECU Software Number Prefix Byte#2	M	ASCII																	
	6	ECU Software Number Byte#0	M	BCD																	
	7	ECU Software Number Byte#1	M	BCD																	
	8	ECU Software Number Byte#2	M	BCD																	
	9	ECU Software Number Byte#3	M	BCD																	
	10	ECU SW Index Number	M	ASCII																	
	11	ECU SW Index Number	M	ASCII																	
	Parameter:																				
	ECU Calibration Part Number from BOM/Teamcenter: e.g.: SOW11223344																				
	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
53	4F	57	12	34	56	78	41	41													
Note:																					
- The Calibration Part Number is defined by the OEM, and shall be embedded in the ECU Calibration file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Vehicle Manufacture ECU Calibration Part Number is stored. - Also see DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level.																					

Specification

4.8 DID 0XF101 Boot Loader Version Number

Service	0x22																											
DID	0xF101																											
Short Description	Read Boot Loader Version number																											
Implementation	Application	Flash Boot Loader																										
	Read(physical and functional addressing)	Read																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for the ECU's with flash functionality																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Boot loader version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Boot loader version number high byte#0- - High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU Boot loader version number high byte#0- Low nibble</td><td>M</td><td>(00-99) BCD</td></tr><tr><td>4</td><td>ECU Boot loader version number high byte#1</td><td>M</td><td>(00-99) BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Boot loader version number [] = [			3	ECU Boot loader version number high byte#0- - High nibble Byte	M	Unsigned	3	ECU Boot loader version number high byte#0- Low nibble	M	(00-99) BCD	4	ECU Boot loader version number high byte#1	M	(00-99) BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Boot loader version number [] = [																										
	3	ECU Boot loader version number high byte#0- - High nibble Byte	M	Unsigned																								
	3	ECU Boot loader version number high byte#0- Low nibble	M	(00-99) BCD																								
	4	ECU Boot loader version number high byte#1	M	(00-99) BCD																								
		]																										
	Parameter: Example Boot loader version number: e.g: V12.34																											
	<table><tr><td>Byte3 High Nibble</td><td>Byte3 Low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 High Nibble	Byte3 Low Nibble	Byte4	1	2	34																		
	Byte3 High Nibble	Byte3 Low Nibble	Byte4																									
	1	2	34																									
	Note: - This identifier shall be defined by supplier and written at the supplier side, and shall not be able to be modified during the vehicle life. - All values 0x00 means no System Supplier Boot Loader Version Number is stored.																											

Specification

4.9 DID 0xF103 Vehicle Manufacturer ECU Purchase Part Number

Service	0x22																					
DID	0xF103																					
Short Description	Vehicle Manufacturer ECU Purchase Part Number																					
Implementation	Application	Flash Boot Loader																				
	Read (physical and functional addressing)	Updated during software download Not readable in Boot Loader																				
Security Level	Read	Write																				
	Level 0	N/A																				
Usage	Mandatory																					
Data	Data Byte No.	Parameter Name	Cvt	Value																		
		ECU Software Part Number [] = []																				
	3	ECU Software Number Prefix Byte#0	M	ASCII																		
	4	ECU Software Number Prefix Byte#1	M	ASCII																		
	5	ECU Software Number Prefix Byte#2	M	ASCII																		
	6	ECU Software Number Byte#0	M	BCD																		
	7	ECU Software Number Byte#1	M	BCD																		
	8	ECU Software Number Byte#2	M	BCD																		
	9	ECU Software Number Byte#3	M	BCD																		
	10	ECU SW Index Number	M	ASCII																		
	11	ECU SW Index Number	M	ASCII																		
		]																				
	Parameter:																					
	ECU Purchase Part Number from BOM/Team centre:																					
	e.g. EEP12345678AA																					
	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41	41
	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
	53	4F	57	12	34	56	78	41	41													
	Note:																					
	- The ECU Purchase Part Number is defined by the OEM and must be written at Supplier manufacturing process. - This DID is used to identify a unique combination of ECU Hardware and reprogrammable Software and Calibration. - If ECU Hardware or Software or Calibration are changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Purchase Part Number or Revision Level <u>must be changed</u> - All Bytes 0x00 means no ECU Purchase Part Number is stored. - Also see DID 0xF143 Vehicle Manufacturer ECU Purchase Part Number Revision Level																					

Specification

4.10 DID 0xF104 Vehicle Manufacturer ECU Application Software Part Number

Service	0x22																				
DID	0xF104																				
Short Description	Application Software Part Number defined by the OEM																				
Implementation	Application		Flash Boot Loader																		
	Read(physical and functional addressing)		Updated during software download Not readable in PBL																		
Security Level	Read		Write																		
	Level 0		N/A																		
Usage	Mandatory for type 1 ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188																				
Data	Data Byte No.	Parameter Name	Cvt	Value																	
		ECU Software Part Number [] = []																			
	3	ECU Software Number Prefix Byte#0	M	ASCII																	
	4	ECU Software Number Prefix Byte#1	M	ASCII																	
	5	ECU Software Number Prefix Byte#2	M	ASCII																	
	6	ECU Software Number Byte#0	M	BCD																	
	7	ECU Software Number Byte#1	M	BCD																	
	8	ECU Software Number Byte#2	M	BCD																	
	9	ECU Software Number Byte#3	M	BCD																	
	10	ECU SW Index Number	M	ASCII																	
	11	ECU SW Index Number	M	ASCII																	
	Parameter:																				
	Application Software Part Number from BOM/Teamcenter e.g. SOW12345678AA																				
<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41	41
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
53	4F	57	12	34	56	78	41	41													
Note:																					
- The Application Software Number is defined by the OEM, and shall be embedded in the ECU Application Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Application Software Number is stored. - Also see DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level																					

Specification

4.11 DID 0xF105 Vehicle Manufacturer ECU Basic Software Part Number

Service	0x22																				
DID	0xF105																				
Short Description	Basic Software Part Number defined by the OEM																				
Implementation	Application	Flash Boot Loader																			
	Read (physical and functional addressing)	Updated during software download Not readable in PBL																			
Security Level	Read	Write																			
	Level 0	N/A																			
Usage	Mandatory for type 1 ECUs with flashing function and whose basic software (BSW) packages can be programmed separately from application software (ASW), else use DID 0xF188.																				
Data	Data Byte No.	Parameter Name	Cvt	Value																	
		ECU Software Part Number [] = []																			
	3	ECU Software Number Prefix Byte#0	M	ASCII																	
	4	ECU Software Number Prefix Byte#1	M	ASCII																	
	5	ECU Software Number Prefix Byte#2	M	ASCII																	
	6	ECU Software Number Byte#0	M	BCD																	
	7	ECU Software Number Byte#1	M	BCD																	
	8	ECU Software Number Byte#2	M	BCD																	
	9	ECU Software Number Byte#3	M	BCD																	
	10	ECU SW Index Number	M	ASCII																	
	11	ECU SW Index Number	M	ASCII																	
	Parameter: Basic Software Part Number from BOM/Teamcenter e.g. SOW12345678AA <table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
53	4F	57	12	34	56	78	41	41													
Note: - The Basic Software Part Number is defined by the OEM, and shall be embedded in the ECU Basic Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Basic Software Part Number is stored. - Also see DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level																					

Specification

4.12 DID 0xF106 Vehicle Manufacturer ECU HMI Software Data Part Number

Service	0x22																				
DID	0xF106																				
Short Description	HMI Software Part Number defined by the OEM																				
Implementation	Application		Flash Boot Loader																		
	Read (physical and functional addressing)		Updated during software download Not readable in PBL																		
Security Level	Read		Write																		
	Level 0		N/A																		
Usage	Mandatory for ECUs with HMI software module and can be reprogrammed.																				
Data	Data Byte No.	Parameter Name	Cvt	Value																	
		ECU Software Part Number [] = []																			
	3	ECU Software Number Prefix Byte#0	M	ASCII																	
	4	ECU Software Number Prefix Byte#1	M	ASCII																	
	5	ECU Software Number Prefix Byte#2	M	ASCII																	
	6	ECU Software Number Byte#0	M	BCD																	
	7	ECU Software Number Byte#1	M	BCD																	
	8	ECU Software Number Byte#2	M	BCD																	
	9	ECU Software Number Byte#3	M	BCD																	
	10	ECU SW Index Number	M	ASCII																	
	11	ECU SW Index Number	M	ASCII																	
		]																			
	Parameter:																				
	HMI Software Part Number from TeamCenter e.g. SOW12345678AA																				
<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	53	4F	57	12	34	56	78	41	41
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11													
53	4F	57	12	34	56	78	41	41													
Note:																					
- This HMI Software Part Number is defined by OEM, and shall be embedded in the ECU HMI file. - This DID will be updated during the reprogramming process. - All Bytes 0x00 means no HMI Software Number is stored - Also see DID 0xF146 Vehicle Manufacturer ECU HMI Software Data Part Number Revision Level																					

Specification

4.13 DID 0xF107 Programming Fingerprint

Service	0x22 / 0x2E																																							
DID	0xF107																																							
Short Description	Programming fingerprint																																							
Implementation	Application		Flash Boot Loader																																					
	Read (physical and functional addressing)		Mandatory Write for Flashable ECU's																																					
Security Level	Read		Write																																					
	Level 0		Level 4(programming session)																																					
Usage	Mandatory for the ECUs with flash functionality																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Write Fingerprint[] = [</td><td></td><td></td></tr><tr><td>3</td><td>programming Date [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>4</td><td>programming Date [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>5</td><td>programming Date [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>6</td><td>repairShopCodeOrTesterSerialNumber[byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td></td><td></td></tr><tr><td>32</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Write Fingerprint[] = [			3	programming Date [byte#1] (YY)	M	BCD(00-99)	4	programming Date [byte#2] (MM)	M	BCD(01-12)	5	programming Date [byte#3] (DD)	M	BCD(01-31)	6	repairShopCodeOrTesterSerialNumber[byte#1] (MSB)	M	ASCII	...				32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Write Fingerprint[] = [																																						
	3	programming Date [byte#1] (YY)	M	BCD(00-99)																																				
	4	programming Date [byte#2] (MM)	M	BCD(01-12)																																				
	5	programming Date [byte#3] (DD)	M	BCD(01-31)																																				
	6	repairShopCodeOrTesterSerialNumber[byte#1] (MSB)	M	ASCII																																				
	...																																							
	32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter: e.g.: 130128 2W...KL																																							
	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>...</td><td>Byte 31</td><td>Byte32</td></tr><tr><td>13</td><td>01</td><td>28</td><td>2</td><td>W</td><td>...</td><td>K</td><td>L</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	...	Byte 31	Byte32	13	01	28	2	W	...	K	L																				
	Byte3	Byte4	Byte5	Byte6	Byte7	...	Byte 31	Byte32																																
	13	01	28	2	W	...	K	L																																
	Note: - Defined by OEM and written in aftersales. - Programming Date is BCD code, and default value is 0x00. - Byte7 to Byte16 represents the Repair Shop Code in ASCII format. - the following 17 bytes represent the tester Serial number, and the tester serial number is reserved. - default value is all bytes space(0x20).																																							
	0xF131 to 0xF135 are the blocked and used to store the programming fingerprint. Programming fingerprint written should use DID 0xF107 by the service 0x2E. The data shall be stored at 0xF131 if the fingerprint data is written for the first time. If another data need to be written (also with 0xF107), block 0xF132 should be filled and so on. Reading 0xF107 shall always return that last programming fingerprint that was written. After the last block of 0xF135 is filled and then if all available block is populated, 0xF132 shall be overwritten, then 0xF133 and so on. 0xF131 should be leaved and can't be overwritten. DID F131 to F135 shall be readable as same as DID F107 including the length and format.																																							

Specification

4.14 DID 0xF111 Factory Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E																																			
DID	0xF111																																			
Short Description	Read / Write factory Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Optional Read / Write	N/A																																		
Security Level	Read	Write																																		
	Level 0	Level 3 (Coding session 10 41)																																		
Usage	Mandatory for any ECU category 1 & 2																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Factory Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Factory Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>4</td><td>Factory Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Factory Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Factory Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Factory Vehicle Order Data VOD [] = [			3	Factory Vehicle Order Data VOD Format Byte	M	Integer	4	Factory Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Factory Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Factory Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Factory Vehicle Order Data VOD [] = [																																		
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	4	Factory Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Factory Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Factory Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5 Dec - Length of the Vehicle Order Data VOD Data = 5 Bytes. - The ECU in which factory VOD stored, shall reserve 255 Bytes for factory VOD.																																			
	Note:																																			
	- This data would be defined by OEM and ECU supplier together, and only can be written once during the manufacturing process at the OEM side. Once written, this can't be changed any more. . - All Bytes FF means no Factory Vehicle Order Data (VOD) is stored - The Factory Vehicle Order Data VOD includes, - the Format Byte 3 will not be included in the CRC calculation, CRC: shall start from byte #4 to n Byte field, which represents the description of the options and topology of the whole vehicle as bit pattern and the CRC is the last Byte. - The Factory Vehicle Order Data VOD is written in two ECUs in the vehicle - For more information please refer the Coding Specification and VOD definition sheet.																																			

Specification

4.15 F11DID 0xF112 Current Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E																																			
DID	0xF112																																			
Short Description	Read / Write Current Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Optional Read / Write	N/A																																		
Security Level	Read	Write																																		
	Level 0	Level 3 (Coding Session 10 41)																																		
Usage	Mandatory for any ECU category 1 & 2																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Current Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Current Vehicle Order Data VOD [] = [			3	Current Vehicle Order Data VOD Format Byte	M	Integer	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Current Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Current Vehicle Order Data VOD [] = [																																		
	3	Current Vehicle Order Data VOD Format Byte	M	Integer																																
	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Current Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5dec - Length of the Vehicle Order Data VOD Data = 5 Byte - The content of this DID is defined and managed by the Project. - The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD																																			
	Note:																																			
- DID would be defined and written by OEM and ECU supplier together. - All Bytes FF means no Factory Vehicle Order Data VOD is stored. - the Format Byte 3 will not be included in the CRC calculation, CRC: shall start from byte #4 to n Byte field, which represents the description of the options and topology of the whole vehicle as bit pattern and the CRC is the last Byte. - Current VOD shall be same as factory VOD if function/features/options of the whole vehicle is never been changed. - Service shall update this Current VOD should a feature/function/option change in service that is impacted by the coding strategy. - For more information, please refer the Coding Specification and VOD definition sheet.																																				

Specification

4.16 DID 0xF108 Coding Data String CDS Data Identifier

Service	0x22 /0x2E			
DID	0xF108			
Short Description	Read / Write Coding Data String CDS			
Implementation	Application	Flash Boot Loader		
	Optional Read / Write	N/A		
Security Level	Read	Write		
	Level 0	Level 3 (coding Session 10 41)		
Usage	Mandatory for the ECUs with CDS Coding functionality			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		Coding Data String CDS[] = [		
	3	CDS Format Byte High Byte#0	M	Integer
	4	CDS Format Byte Low Byte#1	M	Integer
	5	CDS Short VIN High Byte#0	M	ASCII
	...		...	
	11	CDS Short VIN Low Byte#6	M	ASCII
	12	CDS Data High Byte#0	M	Mixed
	...		...	
	n	CDS Data Low Byte#n	M	Mixed
	n+1	CDS CRC	M	Integer
		]		

Specification

4.17 DID 0xF10A Coding for CDS Fingerprint

Service	0x22 /0x2E																																							
DID	0xF10A																																							
Short Description	Coding for CDS fingerprint																																							
Implementation	Application	Flash Boot Loader																																						
	Read / Write	N/A																																						
Security Level	Read	Write																																						
	Level 0	Level 3 (Coding Session 10 41)																																						
Usage	Mandatory for the ECUs with Coding functionality.																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Coding for CDS fingerprint[] = [</td><td></td><td></td></tr><tr><td>3</td><td>codingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>4</td><td>codingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>5</td><td>codingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>6</td><td>repairShopCodeOrTesterSerialNumber [byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>32</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Coding for CDS fingerprint[] = [			3	codingDate [byte#1] (YY)	M	BCD(00-99)	4	codingDate [byte#2] (MM)	M	BCD(01-12)	5	codingDate [byte#3] (DD)	M	BCD(01-31)	6	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII	...		...		32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Coding for CDS fingerprint[] = [																																						
	3	codingDate [byte#1] (YY)	M	BCD(00-99)																																				
	4	codingDate [byte#2] (MM)	M	BCD(01-12)																																				
	5	codingDate [byte#3] (DD)	M	BCD(01-31)																																				
	6	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII																																				
	...		...																																					
	32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter: Coding for CDS fingerprint format: 130128xxxxxxxxxx																																							
	Note: - Defined and written by OEM. - All byte 0 means no CDS fingerprint is stored. - Coding Date is BCD code. - Byte 7 to Byte16 represents the Repair Shop Code in ASCII format. - The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.																																							

Specification

4.18 DID 0xF113 Coding for VOD Fingerprint

Service	0x22 /0x2E																																							
DID	0xF113																																							
Short Description	Coding for VOD fingerprint																																							
Implementation	Application	Flash Boot Loader																																						
	Read / Write	N/A																																						
Security Level	Read	Write																																						
	Level 0	Level 3 (Coding Session 10 41)																																						
Usage	Mandatory for any ECU category 1 & 2																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Coding for VOD fingerprint[] = [</td><td></td><td></td></tr><tr><td>3</td><td>codingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>4</td><td>codingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>5</td><td>codingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>6</td><td>repairShopCodeOrTesterSerialNumber [byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>32</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Coding for VOD fingerprint[] = [			3	codingDate [byte#1] (YY)	M	BCD(00-99)	4	codingDate [byte#2] (MM)	M	BCD(01-12)	5	codingDate [byte#3] (DD)	M	BCD(01-31)	6	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII	...		...		32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Coding for VOD fingerprint[] = [																																						
	3	codingDate [byte#1] (YY)	M	BCD(00-99)																																				
	4	codingDate [byte#2] (MM)	M	BCD(01-12)																																				
	5	codingDate [byte#3] (DD)	M	BCD(01-31)																																				
	6	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII																																				
	...		...																																					
	32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter:																																							
	Coding for VOD fingerprint format: 130128xxxxxxxxxx																																							
	Note:																																							
- Defined and written by OEM. - All byte 0 means no VOD fingerprint is stored. - Coding Date is BCD code. - Byte 6 to Byte16 represents the Repair Shop Code in ASCII format. - The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.																																								

Specification

4.19 DID 0xF10C ECU Body CAN dbc file version.

Service	0x22																											
DID	0xF10C																											
Short Description	ECU body CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all body CAN ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU body CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU body CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU body CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU body CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU body CAN .dbc version [] = [			3	ECU body CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU body CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU body CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU body CAN .dbc version [] = [																										
	3	ECU body CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU body CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU body CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 12.34																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU DBC file version is stored. - All ECU's connected on Body CAN should support this DID.																												

Specification

4.20 DID 0xF11A ECU Powertrain CAN dbc file version

Service	0x22																											
DID	0xF11A																											
Short Description	ECU PT CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all Powertrain CAN ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Powertrain CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU Powertrain CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Powertrain CAN .dbc version [] = [			3	ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte	M	BCD	4	ECU Powertrain CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Powertrain CAN .dbc version [] = [																										
	3	ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte	M	BCD																								
	4	ECU Powertrain CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 12.34																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU DBC file version is stored. - All Powertrain CAN ECU should support this DID.																												

Specification

4.21 DID 0xF11B ECU Chassis CAN dbc file version

Service	0x22																											
DID	0xF11B																											
Short Description	ECU Chassis CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all Chassis CAN ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Chassis CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Chassis CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU Chassis CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU Chassis CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Chassis CAN .dbc version [] = [			3	ECU Chassis CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU Chassis CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU Chassis CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Chassis CAN .dbc version [] = [																										
	3	ECU Chassis CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU Chassis CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU Chassis CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
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	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU DBC file version is stored. - All Chassis CAN ECU should support this DID.																											

Specification

4.22 DID 0xF10E ECU Infotainment CAN dbc file version

Service	0x22																											
DID	0xF10E																											
Short Description	ECU Infotainment CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all Infotainment CAN ECUs																											
Data	<table><thead><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr></thead><tbody><tr><td></td><td>ECU Infotainment CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Infotainment CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU Infotainment CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU Infotainment CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></tbody></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Infotainment CAN .dbc version [] = [			3	ECU Infotainment CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU Infotainment CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU Infotainment CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Infotainment CAN .dbc version [] = [																										
	3	ECU Infotainment CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU Infotainment CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU Infotainment CAN .dbc version high byte#1	M	BCD																								
		]																										
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	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	• Defined by OEM, but written at the supplier side.																											
• All byte 0 means no ECU DBC file version is stored.																												
• All Infotainment CAN ECUs should support this DID.																												

Specification

4.23 DID 0xF125 ECU ADAS CAN dbc file version

Service	0x22																											
DID	0xF125																											
Short Description	ECU connected to ADAS CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for ECU's on ADAS CAN																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU ADAS CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU ADAS CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU ADAS CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU ADAS CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU ADAS CAN .dbc version [] = [			3	ECU ADAS CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU ADAS CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU ADAS CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU ADAS CAN .dbc version [] = [																										
	3	ECU ADAS CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
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	4	ECU ADAS CAN .dbc version high byte#1	M	BCD																								
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	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no Diagnostic DBC file version is stored. - All ADAS CAN ECU should support this DID.																												

Specification

4.24 DID 0xF126 ECU Diagnostic CAN dbc file version

Service	0x22																											
DID	0xF126																											
Short Description	ECU connected to Diagnostic CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for ECU's on Diagnostic CAN																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Diag CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Diag CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU Diag CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU Diag CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Diag CAN .dbc version [] = [			3	ECU Diag CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU Diag CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU Diag CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Diag CAN .dbc version [] = [																										
	3	ECU Diag CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU Diag CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU Diag CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 12.34																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no Diagnostic DBC file version is stored. - All Diagnostic CAN ECU should support this DID.																												

Specification

4.25 DID 0xF127 ECU FA CAN dbc file version

Service	0x22																											
DID	0xF127																											
Short Description	ECU connected to FA CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for ECU's on FA CAN (VF35 ICE only)																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU FA CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU FA CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU FA CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU FA CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU FA CAN .dbc version [] = [			3	ECU FA CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	3	ECU FA CAN .dbc version -high byte#0- low nibble Byte	M	BCD	4	ECU FA CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU FA CAN .dbc version [] = [																										
	3	ECU FA CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU FA CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU FA CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no FA DBC file version is stored. - All FA CAN ECU should support this DID.																											

Specification

4.26 DID 0xF128 ECU FLEXRAY FIBEX file version

Service	0x22			
DID	0xF128			
Short Description	ECU connected to FLEXRAY FIBEX file version			
Implementation	Application	Flash Boot Loader		
	Read (physical and functional addressing)	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Mandatory for ECU's on FLEXRAY bus (VF35 ICE only)			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		ECU Flexray Fibex .dbc version [] = [		
	3	ECU Flexray Fibex file version -high byte#0— High nibble Byte	M	Unsigned
	3	ECU Flexray Fibex file version -high byte#0- low nibble Byte	M	BCD
	4	ECU Flexray Fibex file version high byte#1	M	BCD
		]		
	Parameter:			
	• e.g. 1234			
	Byte3 high Nibble		Byte3 low Nibble	Byte4
	1		2	34
	Note:			
	• Defined by OEM, but written at the supplier side.			
• All byte 0 means no Flexray Fibex file version is stored.				
• All Flexray Fibex FA CAN ECU should support this DID.				

Specification

4.27 DID 0xF10D ECU LIN1 LDF file version

Service	0x22																											
DID	0xF10D																											
Short Description	ECU LIN1 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN1 ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN1 .ldf version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU LIN1 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU LIN1 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU LIN1 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN1 .ldf version [] = [			3	ECU LIN1 .ldf version -high byte#0— High nibble Byte	M	Unsigned	3	ECU LIN1 .ldf version -high byte#0- low nibble Byte	M	BCD	4	ECU LIN1 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN1 .ldf version [] = [																										
	3	ECU LIN1 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU LIN1 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU LIN1 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter's e.g. 1234																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note: - Defined by OEM, but written at the supplier side. - All byte 0 means no ECU LDF file version is stored. - All LIN1 CAN ECU should support this DID.																											

Specification

4.28 DID 0xF11C ECU LIN2 LDF file version

Service	0x22																											
DID	0xF11C																											
Short Description	ECU LIN2 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN2 ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN2 .ldf version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU LIN2 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU LIN2 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU LIN2 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN2 .ldf version [] = [			3	ECU LIN2 .ldf version -high byte#0— High nibble Byte	M	Unsigned	3	ECU LIN2 .ldf version -high byte#0- low nibble Byte	M	BCD	4	ECU LIN2 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN2 .ldf version [] = [																										
	3	ECU LIN2 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU LIN2 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU LIN2 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter's e.g. 1234 <table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note: - Defined by OEM, but written at the supplier side. - All byte 0 means no ECU LDF file version is stored. - All LIN2 CAN ECU should support this DID.																											

Specification

4.29 DID 0xF129 ECU LIN3 LDF file version

Service	0x22																											
DID	0xF129																											
Short Description	ECU LIN3 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN3 ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN3 .ldf version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU LIN3 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU LIN3 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU LIN3 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN3 .ldf version [] = [			3	ECU LIN3 .ldf version -high byte#0— High nibble Byte	M	Unsigned	3	ECU LIN3 .ldf version -high byte#0- low nibble Byte	M	BCD	4	ECU LIN3 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN3 .ldf version [] = [																										
	3	ECU LIN3 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU LIN3 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU LIN3 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no BCM LIN1 LDF file version is stored. - All LIN3 ECU should support this DID.																											

Specification

4.30 DID 0xF10F Service Date

Service	0x22 / 0x2E							
DID	0xF10F							
Short Description	The date of last Service							
Implementation	Application	Flash Boot Loader						
	Read/Write	N/A						
Security Level	Read	Write						
	Level 0	Level 1						
Usage	Mandatory for all ECU's which support Software/Calibration/Content Change							
Data								
	Data Byte No.	Parameter Name	Cvt	Value				
		Last Vehicle servicing performed Date[] = [						
	3	Service Date - Year	M	(00 - 99) BCD				
	4	Service Date - Month	M	(01 - 12) BCD				
	5	Service Date - Day	M	(01 - 31) BCD				
		]						
	Parameter:							
	- This date is written by tester tool at the time of a service ID 0x14 (Physical or Functional addressing). - Any other reasons for writing this DID need to be approved by the VinFast Aftersales Diagnostic and Programming Team							
	e.g: 2018.01.21							
<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td></tr><tr><td>18</td><td>01</td><td>21</td></tr></table>			Byte3	Byte4	Byte5	18	01	21
Byte3	Byte4	Byte5						
18	01	21						
Note:								
The default value is all byte 0x00.								

Specification

4.31 DID 0xFD01 Supplier Specific Part Number Data Identifier

Service	0x22			
DID	0XFD01			
Short Description	Read Supplier specific part number			
Implementation	Application	Flash Boot Loader		
	Read (physical and functional addressing)	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Optional or User Conditional			
Data	Data Byte No.	Parameter Name	Cvt	Value
		Supplier specific part number [] = [		
	3	Supplier Specific Part Number High Byte#0	M	ASCII
	...		...	...
	3+n-1	Supplier Specific Part Number Low Byte#n	M	ASCII
		]		
	Parameter: - It is allowed to use a variable String length. See the component diagnostic information. Note: - This DID will capture any supplier specific part number which needs to be read or reported via 0x22 service, if this is supported then the details of its implementation shall be reported in the ODX.			

Specification

4.32 DID 0xF192 System Supplier ECU Hardware Number

Service	0x22			
DID	0XF192			
Short Description	Read System Supplier hardware part number			
Implementation	Application	Flash Boot Loader		
	Read (physical and functional addressing)	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Optional or User Conditional			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		Supplier specific hardware number [] = [		
	3	Supplier Specific hardware Number High Byte#0	M	ASCII
	...		...	...
	3+n-1	Supplier Specific hardware Number Low Byte#n	M	ASCII
		]		
	Parameter:			
	It is allowed to use a variable String length. See also the component diagnostic information.			
	Note:			
	The system supplier Hardware part number shall be defined and written by the Supplier.			

Specification

4.33 DID 0xF194 System Supplier ECU Software Number

-Service	0x22			
DID	0XF194			
Short Description	Read Supplier specific Software number			
Implementation	Application		Flash Boot Loader	
	Read (physical and functional addressing)		N/A	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Optional or User Conditional			
Data	Data Byte No.	Parameter Name	Cvt	Value
		Supplier specific Software number [] = [		
	3	Supplier Specific Software Number High Byte#0	M	ASCII
	...		...	...
	3+n-1	Supplier Specific Software Number Low Byte#n	M	ASCII
		]		
	Parameter: - It is allowed to use a variable String length. See also the component diagnostic information. Note: - The Supplier specific part number shall be stored under this DID details of its intended use shall be provided in the ODX			

Specification

4.34 DID 0xF110 ECU name data identifier

Service	0x22																											
DID	0XF110																											
Short Description	Read ECU Name																											
Implementation	Application	Flash Boot Loader																										
	Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all ECU's which support diagnostic																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Name [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU Name High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>17</td><td>ECU Name Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Name [] = [			3	ECU Name High Byte#0	M	ASCII	...		...	...	17	ECU Name Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Name [] = [																										
	3	ECU Name High Byte#0	M	ASCII																								
	...		...	...																								
	17	ECU Name Low Byte#n	M	ASCII																								
		]																										
	Parameter: - The max. length of the string is 15 Byte. - The values should be aligned left. - Unused Bytes should be filled with space (0x20)																											
	Note: The ECU will report its name with acronym i.e. BCM = Body Control Module and fill the remaining empty bytes with space ASCII value of space = 0x20Hex so that when its read with a test tool only the ECU name appears BCM																											

Specification

4.35 DID 0xF120 Global Time

Service	0x22																												
DID	0XF120																												
Short Description	Global Time																												
Implementation	Application	Flash Boot Loader																											
	Read (physical and functional addressing)	N/A																											
Security Level	Read	Write																											
	Level 0	N/A																											
Usage	Mandatory for all ECU's which support diagnostic																												
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>6</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	6	M	Unsigned																				
	Size (Bytes)	Cvt	Type																										
	6	M	Unsigned																										
	<table><tr><th>Signal name</th><th>Start byte</th><th>Description</th><th>DID length</th></tr><tr><td rowspan="7">Time stamp</td><td>#1</td><td>0x F120</td><td>48 as defined below</td></tr><tr><td>LSB Byte2</td><td>Date of "Year"</td><td>8</td></tr><tr><td></td><td>Date of "Month"</td><td>8</td></tr><tr><td></td><td>Date of "Day"</td><td>8</td></tr><tr><td></td><td>Date of "Hour"</td><td>8</td></tr><tr><td></td><td>Date of "minute"</td><td>8</td></tr><tr><td></td><td>Date of "second"</td><td>8</td></tr></table>			Signal name	Start byte	Description	DID length	Time stamp	#1	0x F120	48 as defined below	LSB Byte2	Date of "Year"	8		Date of "Month"	8		Date of "Day"	8		Date of "Hour"	8		Date of "minute"	8		Date of "second"	8
	Signal name	Start byte	Description	DID length																									
	Time stamp	#1	0x F120	48 as defined below																									
		LSB Byte2	Date of "Year"	8																									
			Date of "Month"	8																									
			Date of "Day"	8																									
			Date of "Hour"	8																									
			Date of "minute"	8																									
			Date of "second"	8																									
	Parameter: - Reference CAN signal message = MHU_DATE_TIME (0x2F8) - The values should be aligned left. - Unused Bytes should be filled with space (0x00)																												
Note: This data should capture time from CAN and report it out as part of the global snapshot record. ECU can either support 0xF120 or 0xDD07.																													

Specification

4.36 DID 0xF121 Odometer

Service	0x22								
DID	0XF121								
Short Description	Odometer								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>3</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	3	M	Unsigned
	Size (Bytes)	Cvt	Type						
	3	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = IC_MILEAGE/MILE_KM (0x330) - Resolution is 1km/bit. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The Odometer value is transmitted as 3 bytes with resolution of 1bit/Km with maximum value of 16,777,215 Km once this value has reached the ODO meter value shall not increment any further.									

4.37 DID 0xF122 Vehicle Supply Voltage

Service	0x22		
DID	0XF122		
Short Description	Vehicle Supply Voltage		
Implementation	Application	Flash Boot Loader	
	Read (physical and functional addressing)	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic		
Data			
	Size (Bits)	Cvt	Type
	14	M	Unsigned
Parameter:			
- Reference CAN signal message = BCM_VOLTAGE/UBatt (0x10A) - Resolution is 0.000975625V/bit - The values should be aligned left. - Unused Bytes should be filled with space (0x00)			
Note:			
This Voltage value will be on CAN under BCM_VOLTAGE which can be captured into this DID and reported out as a snapshot			

Specification

4.38 DID 0xF123 Ignition status

Service	0x22								
DID	0XF123								
Short Description	Ignition status								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bits)</th><th>Cvt</th><th>Type</th></tr><tr><td>3</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bits)	Cvt	Type	3	M	Unsigned
	Size (Bits)	Cvt	Type						
	3	M	Unsigned						
	Parameter:								
	0x00=OFF 0x01=ACC 0x02=ON 0x03=START 0x04~0x06=reserved 0x07=invalid								
	- Reference CAN signal message = BCM_CLAMP_STAT /STAT_Terminal (0x112) - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
	Note:								
	This DID will indicate the status of the ignition								

4.39 DID 0xF18C ECU SerialNumberDataIdentifier

Service	0x22		
DID	0XF18C		
Short Description	ECUSerialNumberDataIdentifier		
Implementation	Application	Flash Boot Loader	
	Read (physical and functional addressing)	Read (plant)/ Write once only (supplier)	
Security Level	Read	Write	
	Level 0	NA for plant	
Usage	Mandatory for all ECU's which support diagnostic		
Data			
	Size (Bytes)	Cvt	Type
	16	M	ASCII
	Parameter:		
	- The values should be aligned right. - Unused Bytes should be filled with space (0x00)		
	Note:		
	The ECU Serial number should be written ONCE only by the Supplier at their production facilities.		

Specification

4.40 DID 0xDD01 Cabin temperature

Service	0x22								
DID	0XDD01								
Short Description	Cabin temperature								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = CCU_TEMP_PRESSURE/CCU_INTERIORTEMP (0x302) - Supported based on ECU location. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
	Length = 8 Bits								
	Unit = Degree C								
	Value Type = Unsigned								
	Initial Value = 0								
	Factor = 0.5								
Offset = -40									
Min value = -40									
Max value = +85									
Note:									
This will be required to be captured by ECUs which does not have an on board sensor to measure its surrounding temperature and capturing this temperature from the CAN is a lose substitute for the actual temperature and it is based on the location of the ECU, for example an ECU housed in the *trunk (*unless the trunk is a part of the cabin) or in the engine bay need not support the cabin temperature for these ECUs they will need to capture the outside temperature defined in 0xDD02 DID									

Specification

4.41 DID 0xDD02 Outside temperature

Service	0x22								
DID	0XDD02								
Short Description	Outside temperature								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic which are located on outside of the vehicle.								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = IC_TEMP/TEMP_EX_UNFILT(0x2CB) - Supported based on ECU location - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Length = 8 Bits									
Unit = Degree C									
Value Type = Unsigned									
Initial Value = 0									
Factor = 0.5									
Offset = -40									
Min value = -40									
Max value = +85									
Note:									
This will be required to be captured by ECUs which does not have an on board sensor to measure its surrounding temperature and capturing this temperature from the CAN is a lose substitute for the actual temperature and it is based on the location of the ECU. For example, an ECU which is not located outside of vehicle cabin does not need to support this DID.									

Specification

4.42 DID 0xDD03 ECU temperature

Service	0x22		
DID	0XDD03		
Short Description	ECU temperature		
Implementation	Application	Flash Boot Loader	
	Read (physical and functional addressing)	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic		
Data			
	Size (Bytes)	Cvt	Type
	1	M	Unsigned
	Parameter:		
	• The values should be aligned left. • Unused Bytes should be filled with space (0x00)		
	Note:		
	• The ECU Temperature is required to be recorded to be reported as part of the local ECU snapshot where there is no hardware to support this a cabin or outside temperature reading shall be provided to accompany any failure as a substitute. The main reason for this to exclude the environment under which the ECU was operating as a contribution factor.		

4.43 DID 0xDD04 ECU Voltage

Service	0x22								
DID	0XDD04								
Short Description	ECU Voltage								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Resolution is 0.1V/bit - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
- This DID will capture the battery supply voltage at its input pins which will be helpful to compare against the vehicle supply voltage being transmitted on CAN to determine if there is a supply issue causing the fault. - Where the local supply voltage cannot be captured due to hardware limitation, then the ECU shall capture the vehicle supply voltage and associate this with the DTC									

Specification

4.44 DID 0xDD05 Motor RPM

Service	0x22								
DID	0XDD05								
Short Description	Motor RPM								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>4</td><td>O/M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	4	O/M	Unsigned
	Size (Bytes)	Cvt	Type						
	4	O/M	Unsigned						
	Parameter:								
	• Bytes 1-2: Reference CAN signal message = EDS_OUTPUT_FR/EDS_SPD_FR (0x1F8) • Byte 3-4: Reference CAN signal message = EDS_OUTPUT_RR/ EDS_SPD_RR (0x1F9) • Supported only on BEV vehicles. • The values should be aligned Right. • Unused Bytes should be filled with space (0x00)								
	Note:								
	• Where a single motor is fitted only the axle that it is attached to the relevant bytes shall report the motor RPM. i.e. 1-2 for the front and 3-4 for the rear. The Motor RPM shall be directly mapped from the CAN Resolution is 1rpm/bit and offset is -20000 RPM, with range value from -20000 RPM to 20000 RPM.								

4.45 DID 0xDD06 Engine status

Service	0x22		
DID	0XDD06		
Short Description	Engine status		
Implementation	Application	Flash Boot Loader	
	Read (physical and functional addressing)	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic (ICE only)		
Data			
	Size (Bytes)	Cvt	Type
	1	O/M	Bit Mapped
	Parameter:		
	- Supported only on ICE vehicles. - The values should be aligned left.		
Note:			
The ECU name shall be defined by the OEM, but should be written by the Supplier.			

Specification

DID 0xDD07 GPS Time

Service	0x22																												
DID	0XDD07																												
Short Description	GPS Time																												
Implementation	Application	Flash Boot Loader																											
	Read (physical and functional addressing)	N/A																											
Security Level	Read	Write																											
	Level 0	N/A																											
Usage	Mandatory for all ECU's which support diagnostic																												
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>6</td><td>O/M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	6	O/M	Unsigned																				
	Size (Bytes)	Cvt	Type																										
	6	O/M	Unsigned																										
	<table><tr><th>Signal name</th><th>Start byte</th><th>Description</th><th>DID length</th></tr><tr><td rowspan="7">Time stamp</td><td>#1</td><td>0x DD07</td><td>48 as defined below</td></tr><tr><td></td><td>Date of "Year"</td><td>8</td></tr><tr><td></td><td>Date of "Month"</td><td>8</td></tr><tr><td></td><td>Date of "Day"</td><td>8</td></tr><tr><td></td><td>Date of "Hour"</td><td>8</td></tr><tr><td></td><td>Date of "minute"</td><td>8</td></tr><tr><td></td><td>Date of "second"</td><td>8</td></tr></table>			Signal name	Start byte	Description	DID length	Time stamp	#1	0x DD07	48 as defined below		Date of "Year"	8		Date of "Month"	8		Date of "Day"	8		Date of "Hour"	8		Date of "minute"	8		Date of "second"	8
	Signal name	Start byte	Description	DID length																									
	Time stamp	#1	0x DD07	48 as defined below																									
			Date of "Year"	8																									
			Date of "Month"	8																									
			Date of "Day"	8																									
			Date of "Hour"	8																									
		Date of "minute"	8																										
		Date of "second"	8																										
Parameter: • The length of the DID is 6 Byte the year will be 1 byte i.e. 22 as opposed 2022. • Reference CAN signal message = MHU_DATE_TIME (0x2F8) • The values should be aligned left. • Unused Bytes should be filled with space (0x00)																													
Note: • This data should capture time from CAN and report it out as part of the global snapshot record. ECU can either support 0xF120 or 0xDD07 depending upon where they get the signal from.																													

Specification

4.46 DID 0xDD08 Engine RPM

Service	0x22		
DID	0XDD08		
Short Description	Engine RPM		
Implementation	Application	Flash Boot Loader	
	Read (physical and functional addressing)	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic (ICE only)		
Data			
	Size (Bytes)	Cvt	Type
	1	O/M	Unsigned
	Parameter:		
	- Supported only on ICE vehicles. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)		
Note:			
- The ECU name shall be defined by the OEM, but should be written by the Supplier. - Resolution is 50/bit = 12750 with max value of 10000RPM.			

4.47 DID 0xDD09 Vehicle speed

Service	0x22								
DID	0XDD09								
Short Description	Vehicle Speed								
Implementation	Application	Flash Boot Loader							
	Read (physical and functional addressing)	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic only if this is required by the ECU								
Data	<table><tr><th>Size (Bits)</th><th>Cvt</th><th>Type</th></tr><tr><td>13</td><td>M/C</td><td>Unsigned</td></tr></table>			Size (Bits)	Cvt	Type	13	M/C	Unsigned
	Size (Bits)	Cvt	Type						
	13	M/C	Unsigned						
	Parameter:								
	- Reference CAN signal message = IDB_STATUS/VEHICLESPD (0x20D) - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
- Resolution is 13/bits = 460km but is limited 300km/h - Factor = 0.05625.									

Specification

4.48 DID 0xD020 Internal Fail Code

Service	0x22										
DID	0XD020										
Short Description	Internal Fail Code										
Implementation	Application			Flash Boot Loader							
	Read (physical and functional addressing)			N/A							
Security Level	Read			Write							
	Level 0			N/A							
Usage	Optional for all ECU's which support diagnostic										
Data											
	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>105</td><td>O</td><td>Packeted</td></tr></table>					Size (Bytes)	Cvt	Type	105	O	Packeted
	Size (Bytes)	Cvt	Type								
	105	O	Packeted								
	1 byte state encoded			8 bytes Numeric		unsigned numeric					
	State	1 byte		1/2byte Fault Count max count 8 (8 byte)	Fault count	Last Time occurrence (96 bytes)					
	0000	Failure description	CAN transceiver fault	Number of counts per fault	max count 8	yymmddhhmm					
	0001	1	name of device failed	Number of counts per fault	max count 8	6					
	0010	2	name of device failed	Number of counts per fault	max count 8	6					
	0011	3	name of device failed	Number of counts per fault	max count 8	6					
	0100	4	name of device failed	Number of counts per fault	max count 8	6					
	0101	5	name of device failed	Number of counts per fault	max count 8	6					
	0110	6	name of device failed	Number of counts per fault	max count 8	6					
	0111	7	name of device failed	Number of counts per fault	max count 8	6					
	1000	8	name of device failed	Number of counts per fault	max count 8	6					
	1001	9	name of device failed	Number of counts per fault	max count 8	6					
	1010	10	name of device failed	Number of counts per fault	max count 8	6					
	1011	11	name of device failed	Number of counts per fault	max count 8	6					
	1100	12	name of device failed	Number of counts per fault	max count 8	6					
	1101	13	name of device failed	Number of counts per fault	max count 8	6					
	1110	14	name of device failed	Number of counts per fault	max count 8	6					
	1111	15	name of device failed	Number of counts per fault	max count 8	6					
	Parameter:										
	- The values should be aligned left. - Unused Bytes should be filled with space (0x00)										
	Note:										
	The DID data shall be cleared via routine either by the vehicle manufacturer or by the supplier										

Specification

4.49 DID 0x4884 Internal Fault Data

Service	0x22			
DID	0X4884			
Short Description	Internal Fault Data			
Implementation	Application		Flash Boot Loader	
	Read (physical and functional addressing)		N/A	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Optional for all ECU's which support diagnostic			
Data				

Specification

4.50 DID 0xF800 Vehicle Identification Number 2 (VIN2)

Service	0x22			
DID	0XF800			
Short Description	VIN2			
Implementation	Application	Flash Boot Loader		
	Read (physical and functional addressing)	N/A		
Security Level	Read	Write		
	Level 0	Level1&2		
Usage	Mandatory only for all ECU's which support diagnostic where ECU has a calibration which is specific to a vehicle to prevent the ECU being moved to different vehicle where this occurs a misdetection of VIN should flag in the form of calibration invalid and shall continue to display until it is recalibrated. And the ECU shall check the VIN stored in F803 each ignition cycle			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		Vehicle Identification Number [] = [		
	3	Vehicle Identification Number High Byte#0	M/C	ASCII
	...			
	19	Vehicle Identification Number Low Byte#16	M/C	ASCII
		]		
	Parameter:			
	- Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212			
	Note:			
- The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER (0x380) Note this will only report the last 7bytes of the VIN				

Specification

4.51 DID 0xF801 Vehicle Identification Number 3 (VIN3)

Service	0x22																											
DID	0XF801																											
Short Description	VIN3																											
Implementation	Application		Flash Boot Loader																									
	Read		N/A																									
Security Level	Read		Write																									
	Level 0		Level1&2																									
Usage	Mandatory only for all ECU's which support diagnostic where an ECU requires to store VIN to report out as part of DOIP discovery																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M/C</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M/C</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			3	Vehicle Identification Number High Byte#0	M/C	ASCII	...				19	Vehicle Identification Number Low Byte#16	M/C	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	3	Vehicle Identification Number High Byte#0	M/C	ASCII																								
	...																											
	19	Vehicle Identification Number Low Byte#16	M/C	ASCII																								
		]																										
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																											

Specification

4.52 DID 0xF802 Vehicle Identification Number 4 (VIN4)

Service	0x22																											
DID	0XF802																											
Short Description	VIN4																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	Level1&2																										
Usage	Mandatory only for all ECU's which support diagnostic where ECU cannot shared VIN 2 then this VIN should report out VIN as required by SAEJ1979 use case																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M/C</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M/C</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			3	Vehicle Identification Number High Byte#0	M/C	ASCII	...				19	Vehicle Identification Number Low Byte#16	M/C	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	3	Vehicle Identification Number High Byte#0	M/C	ASCII																								
	...																											
	19	Vehicle Identification Number Low Byte#16	M/C	ASCII																								
		]																										
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																											

Specification

4.53 DID 0x0803 Vehicle Identification Number 5 (VIN5)

Service	0x22/2E																										
DID	0X0803																										
Short Description	VIN5																										
Implementation	Application	Flash Boot Loader																									
	Read	N/A																									
Security Level	Read	Write																									
	Level 0	Level1&2																									
Usage	Mandatory for all ECU's which support diagnostic in which VIN needs to be stored as part of Calibration which ties the ECU to a vehicle.																										
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>*M/C</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>*M/C</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>			Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			3	Vehicle Identification Number High Byte#0	*M/C	ASCII	...				19	Vehicle Identification Number Low Byte#16	*M/C	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																							
		Vehicle Identification Number [] = [																									
	3	Vehicle Identification Number High Byte#0	*M/C	ASCII																							
	...																										
	19	Vehicle Identification Number Low Byte#16	*M/C	ASCII																							
		]																									
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																										
	Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																										

*This VIN will only be required to be supported where an ECU has a calibration which has been carried out on vehicle thereby tying the ECU to that specific vehicle if the ECU is moved from the original vehicle to another it should flag a misalignment of calibration requesting calibration be redone with the new vehicle.

Specification

4.54 DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level

Service	0x22					
DID	0xF141					
Short Description	Hardware Number Revision Level of the ECU defined by the OEM.					
Implementation	Application	Flash Boot Loader				
	Read(physical and functional addressing)	Read				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Hardware Part Number Revision Level [] =				
	3	ECU Hardware Number Revision Level	M	BCD		
		]				
	Parameter:					
	- ECU Hardware Part Number Revision Level from BOM/Teamcenter: - This DID represent the Revision Level of ECU Hardware Part Number found in DID 0xF191					
	e.g.: EEP12345678-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- The OEM defines Vehicle Manufacture ECU Hardware Number Revision Level. - The Hardware Number Revision Level must be written by the Supplier. - Byte4 0x00 means no Vehicle Manufacture ECU Hardware Number Revision Level is stored.						

Specification

4.55 DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level

Service	0x22					
DID	0xF142					
Short Description	Calibration Software Number Revision Level of the ECU defined by the OEM					
Implementation	Application	Flash Boot Loader				
	Read	Updated during software download Not readable in Boot Loader				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory for ECUs where Calibration Part Number is to be reported separately					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		Calibration Software Part Number Revision Level [] = [				
	3	ECU Calibration Software Number Release Level	M	BCD		
		]				
	Parameter:					
	- ECU Calibration Software Part Number Release Level from BOM/Teamcenter: - This DID represent the Revision Level of ECU Calibration Software Part Number found in DID 0xF102					
	e.g.: SOW11223344-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:	- The Calibration Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Calibration Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Vehicle Manufacture ECU Calibration Software Number Revision Level is stored.					

Specification

4.56 DID 0xF143 Vehicle Manufacturer Purchase Part Number Revision Level

Service	0x22					
DID	0xF143					
Short Description	Purchase Part Number Revision Level of the ECU defined by the OEM					
Implementation	Application		Flash Boot Loader			
	Read(physical and functional addressing)		Updated during software download Not readable in Boot Loader			
Security Level	Read		Write			
	Level 0		N/A			
Usage	Mandatory					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Part Number Revision Level [] = [				
	4	ECU Number Revision Level	M	BCD		
	Parameter:					
	- ECU Purchase Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Number found in DID 0xF103					
	e.g. EEP12345678-01					
	<table border="1"><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- The ECU Purchase Part Number Revision Level is defined by the OEM and must be written at Supplier manufacturing process. - Byte4 0x00 means no Vehicle Manufacture ECU Hardware Revision Level is stored.						

Specification

4.57 DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level

Service	0x22					
DID	0xF144					
Short Description	Application Software Part Number Revision Level defined by the OEM					
Implementation	Application	Flash Boot Loader				
	Read	Updated during software download Not readable in Boot Loader				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory for ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Application Software Part Number Revision Level [] = [				
	3	ECU Application Software Number Revision Level	M	BCD		
		]				
	Parameter:					
	- Application Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of Application Software Number found in DID 0xF104					
	e.g. SOW12345678-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- The Application Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Application Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Application Software Number Revision Level is stored.						

Specification

4.58 DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level

Service	0x22					
DID	0xF145					
Short Description	Basic Software Part Number Revision Level defined by the OEM					
Implementation	Application		Flash Boot Loader			
	Read		Updated during software download Not readable in Boot Loader			
Security Level	Read		Write			
	Level 0		N/A			
Usage	Mandatory for ECUs which with flashing function and whose basic software can be reprogrammed separately					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Basic Software Part Number Revision Level [] = [				
	4	ECU Basic Software Number Revision Level	M	BCD		
	Parameter:					
	- Basic Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of Basic Software Number found in DID 0xF105					
	e.g. SOW12345678-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- The Basic Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU Basic Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Basic Software Part Number Revision Level is stored.						

Specification

4.59 DID 0xF146 Vehicle Manufacturer ECU HMI Software Part Number Revision Level

Service	0x22					
DID	0xF146					
Short Description	HMI Software Part Number Revision Level defined by the OEM					
Implementation	Application		Flash Boot Loader			
	Read		Updated during software download Not readable in Boot Loader			
Security Level	Read		Write			
	Level 0		N/A			
Usage	Mandatory for ECUs with HMI software module and can be reprogrammed.					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU HMI Software Part Number Revision Level [] = [				
	3	ECU HMI Software Number Revision Level	M	BCD		
		]				
	Parameter:					
	- HMI Software Part Number Revision Level from BOM/Teamcenter. - This DID represents the Revision Level of HMI Software Number found in DID 0xF106					
	e.g. SOW12345678-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- This HMI Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU HMI Software file. - This DID will be updated during the reprogramming process. - Byte4 0x00 means no HMI Software Number Revision Level is stored.						

Specification

4.60 DID 0xF148 Vehicle Manufacturer ECU Software Part Number Revision Level

Service	0x22						
DID	0xF148						
Short Description	Software Part Number Revision Level of the ECU defined by the OEM.						
Implementation	Application		Flash Boot Loader				
	Read(physical and functional addressing)		Updated during software download Not readable in Boot Loader				
Security Level	Read		Write				
	Level 0		N/A				
Usage	Mandatory for all ECUs.						
Data	Data Byte No.	Parameter Name	Cvt	Value			
		ECU Software Part Number Revision Level [] = [					
	4	ECU Software Number Revision Level	M	BCD			
		]					
Parameter:							
- ECU Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Software Part Number found in DID 0xF188							
e.g.: SOW12345678-01							
<table border="1"><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>						Byte3	01
Byte3							
01							
Note:							
- The OEM defines Vehicle Manufacture ECU Software Number Revision Level, and it shall be embedded in the ECU Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Vehicle Manufacture ECU Software Revision Number is stored.							

Specification

4.61 DID 0x016C HV Status and Charging Status

Service	0x22		
DID	0X016C		
Short Description	HV Status and Charging Status		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for VCU only		
Data	Size (Bytes)	Cvt	Type
	1	M	State Encoded
	Parameters:		
	4) HV Status: (0000 00xx)		
	▪ 0x00= HV OFF		
	▪ 0x01=HV ON		
	3) HV Charging Status: (0000 xx00)		
	▪ 0x00= Charging OFF		
	▪ 0x01= Charging ON		
	• The size of each parameter = two bits in length		
• Unused Bytes parameters 1 and 2 should be filled with (0x00)			

Specification

4.62 DID 0xD06B Signed Software Consistency Check

Service	0x22								
DID	0XD06B								
Short Description	Signed Software Consistency Check								
Implementation	Application	Flash Boot Loader							
	Read	Read							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic and is supporting signed software								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>State encoded</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	State encoded
	Size (Bytes)	Cvt	Type						
	1	M	State encoded						
	Parameters: 00 = Consistency not checked 01 = Consistency checked & ok 02 = Signature doesn't match 03 = Error and detail of the error								
	- The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
	Note: This DID is used as part of cyber security to validate signature along with software consistency check								

Specification

4.63 DID 0xD05B Software Status Check

Service	0x22																						
DID	0xD05B																						
Short Description	Software Part Number Status Check																						
Implementation	Application	Flash Boot Loader																					
	Read (physical)	Updated during software download. readable in Boot Loader																					
Security Level	Read	Write																					
	Level 0	N/A																					
Usage	*Mandatory for all ECUs which support dual bank architecture.																						
Data	Data Byte No.	Parameter Name	Cvt	Value																			
		ECU Software Part Number [] = []																					
	3	ECU Software Number Prefix Byte#0	*M/C	ASCII																			
	4	ECU Software Number Prefix Byte#1	*M/C	ASCII																			
	5	ECU Software Number Prefix Byte#2	*M/C	ASCII																			
	6	ECU Software Number Byte#0	*M/C	BCD																			
	7	ECU Software Number Byte#1	*M/C	BCD																			
	8	ECU Software Number Byte#2	*M/C	BCD																			
	9	ECU Software Number Byte#3	*M/C	BCD																			
	10	ECU SW Index Number	*M/C	ASCII																			
	11	ECU SW Index Number	*M/C	ASCII																			
	12	ECU Software Number Byte#4	*M/C	State Encoded																			
	Parameter:																						
	- Byte 3 to Byte 09 = ECU Software Part Number from BOM/Teamcenter - Byte 12: 1 = accepted; 0 = rejected e.g.: SOW12345678AA00																						
<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td><td>Byte11</td><td>Byte12</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td><td>41</td><td>41</td><td>00</td></tr></table>				Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	53	4F	57	12	34	56	78	41	41	00
Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12														
53	4F	57	12	34	56	78	41	41	00														
Note:																							
- The OEM defines Vehicle Manufacture ECU Software Part Number, and it shall be embedded in the ECU Software file. - All Bytes 0x00 means no Vehicle Manufacture ECU Software Part Number is stored.																							

Specification

4.64 DID 0x0222 Valid Application

Service	0x22								
DID	0X0222								
Short Description	Valid Application								
Implementation	Application	Flash Boot Loader							
	Read	Read							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic and is supporting signed software								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>State encoded</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	State encoded
	Size (Bytes)	Cvt	Type						
	1	M	State encoded						
	Parameters: 00 = Not Valid 01 = Valid - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note: This DID is used to store results from Routine 0x0222 Validate Application									

Specification

4.65 DID 0xF16A Current Vehicle Order Data VOD Data Identifier Cat 1 ECU

Service	0x22 /0x2E																																			
DID	0xF16A																																			
Short Description	Read / Write Current Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Optional Read / Write	N/A																																		
Security Level	Read	Write																																		
	Level 0	L3																																		
Usage	Mandatory for any ECU category 1 which supports VOD																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Current Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Current Vehicle Order Data VOD [] = [			3	Current Vehicle Order Data VOD Format Byte	M	Integer	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Current Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Current Vehicle Order Data VOD [] = [																																		
	3	Current Vehicle Order Data VOD Format Byte	M	Integer																																
	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Current Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5dec - Length of the Vehicle Order Data VOD Data = 5 Byte - The content of this DID is defined and managed by the Project. - The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD																																			
	Note:																																			
	- DID would be defined and written by OEM and ECU supplier together. - All Bytes FF means no Vehicle Order Data VOD is stored. - This DID shall store all of the VOD data to which the ECU subscribes to. - Service shall update/code this DID upon the requirement of what the ECU needs to supports as part of the software adaptation. - All parameter values which the ECU supports shall be listed in the ODX of the ECU. - For more information, please refer the Coding Specification and VOD definition sheet.																																			

Specification

4.66 DID 0xF16B Category 1 ECU Current and last stored checksum

Service	0x22																									
DID	0XF16B																									
Short Description	VOD Checksum																									
Implementation	Application	Flash Boot Loader																								
	Read	N/A																								
Security Level	Read	Write																								
	Level 0	N/A																								
Usage	Mandatory for all Category 1 ECU's which support VOD																									
Data																										
	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>5</td><td>M</td><td>Hex</td></tr></table>			Size (Bytes)	Cvt	Type	5	M	Hex																	
	Size (Bytes)	Cvt	Type																							
	5	M	Hex																							
	<table><tr><th>Signal name</th><th>Start byte</th><th>Description</th><th>DID length</th></tr><tr><td rowspan="6">Time stamp</td><td>#1</td><td>0x F16B</td><td>40</td></tr><tr><td></td><td>CRC</td><td>8</td></tr><tr><td></td><td>Checksum</td><td>8</td></tr><tr><td></td><td>Checksum</td><td>8</td></tr><tr><td></td><td>Checksum</td><td>8</td></tr><tr><td></td><td>Checksum</td><td>8</td></tr></table>			Signal name	Start byte	Description	DID length	Time stamp	#1	0x F16B	40		CRC	8		Checksum	8		Checksum	8		Checksum	8		Checksum	8
	Signal name	Start byte	Description	DID length																						
	Time stamp	#1	0x F16B	40																						
			CRC	8																						
			Checksum	8																						
		Checksum	8																							
		Checksum	8																							
		Checksum	8																							
Parameter: i.e. 2ABC571234																										
<table><tr><td>Byte1</td><td>Byte2</td><td>Byte3</td><td>Byte4</td><td>Byte5</td></tr><tr><td>2A</td><td>BC</td><td>57</td><td>12</td><td>34</td></tr></table>			Byte1	Byte2	Byte3	Byte4	Byte5	2A	BC	57	12	34														
Byte1	Byte2	Byte3	Byte4	Byte5																						
2A	BC	57	12	34																						
- The values should be aligned left. - Unused Bytes should be filled with space (0x00)																										
Note: - This DID shall capture the checksum of the parameter of the value to which the ECU subscribes to and to which the ECU is configured. The values it compares to are received over CAN from the Master and it stores these locally into 0xF112 DID.																										

Specification

4.67 DID 0xF16C Category 1 ECU Current VOD checksum

Service	0x22		
DID	0XF16C		
Short Description	Current VOD Checksum		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all Category 1 ECU's which support VOD		
Data			

Specification

4.68 DID 0xF16D VOD Validation Flag for Category 1 and 2 ECUs

Service	0x22								
DID	0XF16D								
Short Description	States of ECU VOD validation flag								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support VOD requirement.								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>State encoded</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	State encoded
	Size (Bytes)	Cvt	Type						
	1	M	State encoded						
	State: 00 = Current Configuration Invalid								
	State: 01 = Current Configuration Valid.								
	- The values should be aligned left. - Unused Bits shall not be displayed.								
	Note:								
	- This DID shall report the Validation of the current VOD that the ECU has configured to. This shall be supported by both Category 1 and 2 ECU, each time a new parameter is required to be absorbed the validation flag shall be set to invalid as soon as VOD is validated then the flag shall be set to valid, used by all ECUs updating their configuration. - The master ECU after it has validated the VOD internally shall use this as a trigger to transmit the VOD.								

Specification

4.69 DID 0xEF00 VOD Validation States for a Category 2 ECU

Service	0x22		
DID	0XEF00		
Short Description	States of ECU validation		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support VOD requirement.		
Data	Size (Bytes)	Cvt	Type
	1	M	State encoded
State: 00 = Current configuration compared and same as factory State: 01 = Reading factory configuration DID F111 State: 02 = Reading Current Configuration DID F112 State: 03 = Failed Factory VOD not configured DTC U1501-55 State: 04 = Comparing current configuration State: 05 = Last Stored configuration DID EF02 State: 06 = Comparing current and last stored configuration State: 07 = Current and Stored Configuration Compared and OK State: 08 = Learning New configuration State: 09 = Failed to learn Configuration U1502-51(Not Programmed) State: 0A = Validating Configuration State: 0B = Conflict Between Values of Configuration Parameters U1502-F1 State: 0C = Installed Hardware Mismatch with Configuration Parameters U1502-F0 State: 0D = Applying Configuration Update State: 0E = Software Execution Error U1502-94 component or system operating conditions State: 0F = Configuration Not Supported by Software U1502-57 State: 10 = Failed to Store DTC U1502-46 Calibration / Parameter Memory Failure State: 11 = Failed Current VOD not configured DTC U1502-55 State: 12 = Configure current VOD State: 13 = Configure Factory VOD State: 14 = Set Configuration as Valid State: 15 = Set Configuration as Invalid State: 16 = Transmit VOD on CAN State: 17 = Failed to compare DTC U1502-F2 • The values should be aligned left. • Unused Bits shall not be displayed. Note: • This DID shall report all of the VOD validation States of Category 2 ECU			

Specification

4.70 DID 0xEF01 Category 2 ECU compared checksum

Service	0x22		
DID	0XEF01		
Short Description	VOD Compared Checksum		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all Category 2 ECU's which support VOD		
Data			

Size (Bytes)	Cvt	Type
1	M	Hex

Signal name	Start byte	Description	DID length
Time stamp	#1	0x EF01	40
		CRC	8

Parameter:

- Reference CAN signal message = VOD2_Checksum
- The values should be aligned left.
- Unused Bytes should be filled with space (0x00)

Note:

- This DID shall be used by the Master and Backup ECUs to store the Validation Checksum to be used by the backup to compare its local version of the VOD payload against what it receives from the Master on CAN. The Master after it has validated the VOD payload shall store it and broadcast it on CAN for other ECU to use and validate their local stored payload which they have acquired from CAN.

Specification

4.71 DID 0xEF02 Last Stored Vehicle Order Data VOD Data Cat 2 ECU

Service	0x22 /0x2E																																			
DID	0xEF02																																			
Short Description	Last stored Current Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Read	N/A																																		
Security Level	Read	Write																																		
	Level 0	NA																																		
Usage	Mandatory for Master and Backup ECUs category 2																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Current Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Current Vehicle Order Data VOD [] = [			3	Current Vehicle Order Data VOD Format Byte	M	Integer	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Current Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Current Vehicle Order Data VOD [] = [																																		
	3	Current Vehicle Order Data VOD Format Byte	M	Integer																																
	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Current Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5dec - Length of the Vehicle Order Data VOD Data = 5 Byte - The content of this DID is defined and managed by the Project. - The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD																																			
	Note:																																			
	- DID would be defined and written by OEM and ECU supplier together. - All Bytes FF means no Vehicle Order Data VOD is stored. - This DID shall store the last good configuration of the VOD for Category 2 ECU that has passed all of the validation checks. - Whenever a new parameter is updated it shall be learned and if it passes the validation check it shall be stored in this DID for comparison when ECU is powered on subsequently with the current DID 0xF112 to check and ensure that no additional parameters have been changed or updated. - For more information, please refer the Coding Specification and VOD definition sheet.																																			

Specification

4.72 DID 0xEF03 Last Stored Vehicle Order Data VOD Data Cat 2 ECU

Service	0x22 /0x2E																																			
DID	0xEF03																																			
Short Description	Last stored Current Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Read	N/A																																		
Security Level	Read	Write																																		
	Level 0	NA																																		
Usage	Mandatory for any ECU category 1 which support VOD configuration																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Current Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Current Vehicle Order Data VOD [] = [			3	Current Vehicle Order Data VOD Format Byte	M	Integer	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Current Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Current Vehicle Order Data VOD [] = [																																		
	3	Current Vehicle Order Data VOD Format Byte	M	Integer																																
	4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Current Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5dec - Length of the Vehicle Order Data VOD Data = 5 Byte - The content of this DID is defined and managed by the Project. - The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD																																			
	Note:																																			
	- DID would be defined and written by OEM and ECU supplier together. - All Bytes FF means no Vehicle Order Data VOD is stored. - This DID shall store the last good configuration of the VOD for Category 1 ECU that has passed all of the validation checks. - Whenever a new parameter is updated it shall be learned and if it passes the validation check it shall be stored in this DID for comparison when ECU is powered on subsequently with the current DID 0xF112 to check and ensure that no additional parameters have been changed or updated. - For more information, please refer the Coding Specification and VOD definition sheet.																																			

Specification

4.73 DID 0xEF04 ECU Validation States for a Category 1

Service	0x22		
DID	0XEF04		
Short Description	States of ECU validation		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support VOD requirement.		
Data	Size (Bytes)	Cvt	Type
	1	M	State encoded
State: 00 = Reading VOD Current (DID F112) State: 01 = Comparing Current & stored configuration State: 02 = Current and Stored Configuration Compared and OK store checksum in DID F16B State: 03 = Comparing current Configured and last stored configuration State: 04 = Failed Current VOD and stored does not compare DTC U1052-F2 Configuration lost or changed State: 05 = Learning New configuration State: 06 = Failed to learn Configuration U1052-51 (Not Programmed) Configuration State: 07 = Validating Configuration State: 08 = Conflict Between Values of Configuration Parameters TC U1052-F0 State: 09 = Installed Hardware Mismatch with Configuration Parameters DTC U1052-F1 State: 0A = Applying Configuration Update State: 0B = Software Execution Error DTC U1052-94 State: 0C = Configuration Not Supported by Software DTC U1052-57 State: 0D = Last Stored configuration DID EF03 State: 0E = Failed to Store DTC U1052-46 (Calibration / Parameter Memory Failure) State: 0F = Current configured and Stored Configuration Compared and OK store checksum in DID F16B State: 10 = Failed Current VOD Not configured DTCU1052-55 State: 11 = Update Current Configured VOD State: 12 = Failed Current VOD not configured DTC U1051-55 State: 13 = Current VOD not received on CAN State: 14 = Set Configuration as Valid State: 15 = Set Configuration as Invalid • The values should be aligned left. • Unused Bits shall not be displayed. Note: • This DID shall report all of the VOD validation States of Category 1 ECU			

Specification

4.74 DID 0xEF05 Compare Category 2 master and backup ECUs Checksum

Service	0x22		
DID	0XEF05		
Short Description	States of the Master and Backup ECU Payload checksum		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support VOD requirement.		
Data	Size (Bytes)	Cvt	Type
	1	M	State encoded
State: 00 = Current VOD compared is NOT OK State: 01 = Current VOD compared is OK. • The values should be aligned left. • Unused Bits shall not be displayed. Note: • This DID is required to be stored by the backup is used to store the state of the comparison of the checksum between the master and the backup ECU. In case there is a mismatch between the master ECU and Backup ECU to be read with service \$22 in case there is discrepancy the checksum of the Master shall be deemed as correct.			

Specification

4.75 DID 0XF109 HUD FONT Data Block Part Number

Service	0x22						
DID	0xF109						
Short Description	HUD FONT Data Block Part Number defined by the OEM						
Implementation	Application	Flash Boot Loader					
	Mandatory Read	Updated during software download Not readable in Boot Loader					
Security Level	Read	Write					
	Level 0	N/A					
Usage	Mandatory for ECUs Which have large calibration files						
	Data Byte No.	Parameter Name	Cvt	Value			
		Calibration Software Part Number [] = [					
	3	ECU Calibration Number Prefix Byte#0	M	ASCII			
	4	ECU Calibration Number Prefix Byte#1	M	ASCII			
	5	ECU Calibration Number Prefix Byte#2	M	ASCII			
	6	ECU Calibration Number Byte#0	M	BCD			
	7	ECU Calibration Number Byte#1	M	BCD			
	8	ECU Calibration Number Byte#2	M	BCD			
	9	ECU Calibration Number Byte#3	M	BCD			
		]					
	Parameter:						
	ECU HUD FONT Part Number from BOM/Teamcenter: e.g.: SOW11223344						
	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
	53	4F	57	11	22	33	44
	Note:						
	- The HUD data block Part Number is defined by the OEM, and shall be embedded in the ECU Calibration file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Vehicle Manufacture ECU Calibration Part Number is stored. - Also see DID 0xF149 Vehicle Manufacturer ECU HUD Data block Part Number Revision Level.						

Specification

4.76 DID 0xF149 HUD FONT Data Block Part Number Revision Level

Service	0x22					
DID	0xF149					
Short Description	HUD FONT Data Block Part Number Revision Level of the ECU defined by the OEM					
Implementation	Application	Flash Boot Loader				
	Mandatory Read	Updated during software download Not readable in Boot Loader				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory for ECUs Which have large calibration files					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		Calibration Software Part Number Revision Level [] = [				
	3	ECU Calibration Software Number Release Level	M	BCD		
		]				
	Parameter:					
	- ECU HUD FONT part number Release Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Calibration Software Part Number found in DID 0xF109					
	e.g.: SOW11223344-01					
	<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>				Byte3	01
	Byte3					
01						
Note:						
- The HUD FONT Number Revision Level is defined by the OEM, and shall be embedded in the ECU Calibration Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Vehicle Manufacture ECU Calibration Software Number Revision Level is stored.						

Specification

4.77 DID 0x0804 Diagnostic Specification Number Applied

Service	0x22																											
DID	0X0804																											
Short Description	Read Diagnostic specification number																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for All ECUs which use UDS diagnostic requirement																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>3+n-1</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Diagnostic Specification Version number [] = [			3	Diagnostic Specification Version number used Byte#0	M	ASCII	...		...	...	3+n-1	Diagnostic Specification Version number used Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Diagnostic Specification Version number [] = [																										
	3	Diagnostic Specification Version number used Byte#0	M	ASCII																								
	...		...	...																								
	3+n-1	Diagnostic Specification Version number used Low Byte#n	M	ASCII																								
		]																										
	Parameter:																											
	It is allowed to use a variable String length. See also the component diagnostic information. i.e.Vxxx = V1.4 56 01 2E 04																											
	Note:																											
The Diagnostic specification version number used in development of the ECU shall be written by the Supplier.																												

Specification

4.78 DID 0x0805 Flash Specification Number Applied

Service	0x22																											
DID	0X0805																											
Short Description	Read Flash specification number																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for All ECUs which are Flashable																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Flash Specification Version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Flash Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>3+n-1</td><td>Flash Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Flash Specification Version number [] = [			3	Flash Specification Version number used Byte#0	M	ASCII	...		...	...	3+n-1	Flash Specification Version number used Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Flash Specification Version number [] = [																										
	3	Flash Specification Version number used Byte#0	M	ASCII																								
	...		...	...																								
	3+n-1	Flash Specification Version number used Low Byte#n	M	ASCII																								
		]																										
	Parameter: It is allowed to use a variable String length. See also the component diagnostic information. i.e.Vxxx = V1.4 56 01 2E 04																											
	Note: The Flash specification version number used in development of the ECU shall be written by the Supplier.																											

4.79 DID 0x2070 Transport mode

Service	0x22		
DID	0X2070		
Short Description	Status of Transport Mode		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Only Applicable to the BCM.		
Data	Size (Bytes)	Cvt	Type
	1	M	State encoded
	State: 00 = Transport Mode Not Activated		
	State: 01 = Transport Mode Activated.		
	State: 02 = Failed to activate Transport mode		
	- The values should be aligned left. - Unused Bits shall not be displayed.		
	Note:		
	Transport shall be activated via a routine 9010 and this routine shall write to DID 2070 the status of transport mode		

Specification

4.80 DID 0x0179 Check ECU Control of HV

Service	0x22		
DID	0x0179		
Short Description	To Check ECU control of HV Contactor		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for VCU only		
Data	Size (Bytes)	Cvt	Type
	1	M	State Encoded
	Parameters: 1) ECU Control Status: (xx) 0x00= ECU has Control 0x01= Routine 0xF001 has control - This check will be helpful to determine who is controlling the contactor ECU or Routine 0xF001		

4.81 DID 0x0806 AUTOSAR Version Number Applied

Service	0x22			
DID	0X0806			
Short Description	Read Autosar Version number			
Implementation	Application		Flash Boot Loader	
	Read		N/A	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Optional for All ECUs which use UDS diagnostic requirement and Autosar compliant			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		Diagnostic Specification Version number [] = [		
	3	Diagnostic Specification Version number used Byte#0	M	ASCII
	...		...	...
	3+n-1 (max 24 bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII
		]		
	Parameter:			
	It is allowed to use a variable String length. See also the component diagnostic information. i.e.Vxxx = V1.4 56 01 2E 04			
Note:				
The Autosar version number used in development of the ECU shall be written by the Supplier.				

Specification

4.82 DID 0x0807 ODX Template Version Applied

Service	0x22																											
DID	0X0807																											
Short Description	ODX																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Optional for All ECUs which use UDS diagnostic requirement and Autosar compliant																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Diagnostic Specification Version number [] = [			3	Diagnostic Specification Version number used Byte#0	M	ASCII	...		...	...	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Diagnostic Specification Version number [] = [																										
	3	Diagnostic Specification Version number used Byte#0	M	ASCII																								
	...		...	...																								
	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII																								
		]																										
	Parameter:																											
	It is allowed to use a variable String length. See also the component diagnostic information. i.e.Vxxx = V1.4 56 01 2E 04																											
	Note:																											
The ODX template version number used in development of the ECU shall be written by the Supplier.																												

Specification

4.83 DID 0x0808 EOL Template Version Applied

Service	0x22																											
DID	0X0808																											
Short Description	End Of Line (EOL)																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Optional for All ECUs which use UDS diagnostic requirement and Autosar compliant																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Diagnostic Specification Version number [] = [			3	Diagnostic Specification Version number used Byte#0	M	ASCII	...		...	...	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Diagnostic Specification Version number [] = [																										
	3	Diagnostic Specification Version number used Byte#0	M	ASCII																								
	...		...	...																								
	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII																								
		]																										
	Parameter:																											
	It is allowed to use a variable String length. See also the component diagnostic information. i.e.Vxxx = V1.4 56 01 2E 04																											
	Note:																											
The ODX template version number used in development of the ECU shall be written by the Supplier.																												

4.84 DID 0x0809 AS Template Version Applied

Service	0x22																											
DID	0X0809																											
Short Description	After Sales (AS) Template																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Optional for All ECUs which use UDS diagnostic requirement and Autosar compliant																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr><tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Diagnostic Specification Version number [] = [			3	Diagnostic Specification Version number used Byte#0	M	ASCII	...		...	...	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Diagnostic Specification Version number [] = [																										
	3	Diagnostic Specification Version number used Byte#0	M	ASCII																								
	...		...	...																								
	3+n-1 (Max 24bytes)	Diagnostic Specification Version number used Low Byte#n	M	ASCII																								
		]																										

Specification

	Parameter: - It is allowed to use a variable String length. See also the component diagnostic information. i.e. Vxxx = V1.4 56 01 2E 04 Note: - The ODX template version number used in development of the ECU shall be written by the Supplier.
--	--

4.85 DID 0xF13A ECU LIN4 LDF file version

Service	0x22																											
DID	0xF13A																											
Short Description	ECU LIN4 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Mandatory Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN4 ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN4 .ldf version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU LIN4 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU LIN4 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU LIN4 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN4 .ldf version [] = [			3	ECU LIN4 .ldf version -high byte#0— High nibble Byte	M	Unsigned	3	ECU LIN4 .ldf version -high byte#0- low nibble Byte	M	BCD	4	ECU LIN4 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN4 .ldf version [] = [																										
	3	ECU LIN4 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU LIN4 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU LIN4 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter's																											
	• e.g. 1234																											
	<table><tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte3 high Nibble	Byte3 low Nibble	Byte4	1	2	34																		
	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	• Defined by OEM, but written at the supplier side.																											
	• All byte 0 means no ECU LDF file version is stored.																											
• All LIN4 CAN ECU should support this DID.																												

Specification

4.86 DID 0xF13A ECU LIN5 LDF file version

Service	0x22																											
DID	0xF13B																											
Short Description	ECU LIN5 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Mandatory Read (physical and functional addressing)	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN4 ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN5 .ldf version [] = [</td><td></td><td></td></tr><tr><td>3</td><td>ECU LIN5 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>3</td><td>ECU LIN5 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>4</td><td>ECU LIN5 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN5 .ldf version [] = [			3	ECU LIN5 .ldf version -high byte#0— High nibble Byte	M	Unsigned	3	ECU LIN5 .ldf version -high byte#0- low nibble Byte	M	BCD	4	ECU LIN5 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN5 .ldf version [] = [																										
	3	ECU LIN5 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	3	ECU LIN5 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	4	ECU LIN5 .ldf version high byte#1	M	BCD																								
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	e.g. 1234																											
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	Byte3 high Nibble	Byte3 low Nibble	Byte4																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU LDF file version is stored. - All LIN5 CAN ECU should support this DID.																											

Specification

4.87 DID 0x080A Cyber Security Supported

Service	0x22			
DID	0X080A			
Short Description	CS Compliant ECU			
Implementation	Application	Flash Boot Loader		
	Read	Read		
Security Level	Read	Write		
	Level 0	Written by supplier at end of production or as part of SW update		
Usage	Mandatory for All ECUs which are supporting Cyber security			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
	3	Cyber security supported	M	SED
		]		
	Parameter:			
	Size (Bytes)	Cvt	Type	
	1	M	State Encoded	
			0x00 Supported	
			0x01 Not supported	
Note:				
To be supported by all compliant and non-compliant ECUs. This will be read by the diagnostic tool to determine which flash path to follow				

Specification

4.88 DID 0x080B Secure CAN Supported

Service	0x22/2E			
DID	0X080B			
Short Description	Secure CAN Communication			
Implementation	Application		Flash Boot Loader	
	Read/Write		NA	
Security Level	Read		Write	
	Level 1/2		Yes	
Usage	Conditional for ECUs which are supporting Secure CAN			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
	3	Cyber security supported	C/M	SED
		]		
	Parameter:			
	Size (Bytes)	Cvt	Type	
	1	M	SED	
			0x00 ON	
			0x01 OFF	
Note:				
To be supported by all Secure CAN compliant ECUs.				

Specification

5 Standards

5.1 Additional Documents

Sr. No	Name	Version
[1]	Basic_Diagnostic_Specification	latest
[2]	Flash_Specification	latest
[3]	Coding_Specification	latest
[4]	ECU_Identifier_Specification	latest

Table 4: Referenced Documents

Specification

5.2 Norms and Standards

Ref. No	Document Name
ISO14229	Road vehicles — Unified diagnostic services (UDS)
ISO15031	Road vehicles — Communication between vehicle and external equipment for emissions- related Diagnostic
ISO15765	Road vehicles — Diagnostic on Controller Area Networks (CAN)
ISO17987	Road vehicles — Local Interconnect Network LIN
ISO22900	Road vehicles — Modular vehicle communication interface
ISO11898	Road vehicles – Controlled Area Network (CAN)
ISO22901	Road vehicles — Open diagnostic data exchange (ODX)
SAE J1979	E/E Diagnostic Test Modes
SAE J2012	Diagnostic Trouble Code Definitions

Table 5 : Normative references

This Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

6 Lists

6.1 List of tables

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